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Student Name: Nisha Subedi
London Met ID: 23049293
College ID: np05cp4a230236@iic.edu.np
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Internal Supervisor: Mr. Sujan Subedi
External Supervisor: Mr. Utsav Dhungana

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Abstract

It is the introduction of an AI-Powered Crop Disease Detection and Recommendation System that will help farmers to diagnose and manage crop diseases in an efficient, accurate, and timely way. One of the most significant areas in regards to food security and economic growth is agriculture. Nevertheless, crop diseases are an issue of significant concern since they considerably decrease the productivity of agriculture and cause high losses annually. In most developing countries, farmers extensively use the manual field observation, consultation services of agricultural specialists or physical inspection of crops in order to detect diseases. These methods are long-standing, although frequently slow, inconsistent, and inaccurate because of the human factor and unavailability of experts at the time, particularly in rural (or resource-limited) settings.

In a bid to overcome these constraints, the proposed system proposes an automated and intelligent system based on the Artificial Intelligence (AI) and Deep Learning technologies. This system is mainly built around Convolutional Neural Networks (CNN), which is one of the most powerful deep learning models that are highly utilized in image recognition and classification problems. CNN can be used to extract meaningful features of images and detect intricate patterns of crop leaf images which can be used to identify various forms of diseases.

Farmers are able to just post pictures of infected or suspected crop leaves using a user-friendly web interface. After loading the image, the system resorts to the trained CNN model that analyzes visual patterns and makes predictions of the potential diseases in the crop. The system will then produce the detailed results like the name of the predicted disease, the level of confidence of the prediction, the severity of the infection, and the action that should be taken to prevent the disease or what treatment should be taken. This detailed output assists the farmers in not only diagnosing the problem but also making quick and informed decisions to minimise further damage to the crop and enhance the quality of yield.

The system is built based on the Scrum methodology that is an iterative and agile approach to software development. This methodology guarantees a continuous improvement process, proper teamwork and adaptability to the changing requirements during the development process. MERN stack is used to implement the web application, which consists of MongoDB to manage the database, Express.js and Node.js to develop the backend, and React.js to create a responsive and interactive user interface. The deep learning libraries are built on Python like TensorFlow or PyTorch and used to create the machine learning model.

The combination of AI and modern web technologies contributes to the fact that the system is very user-friendly, scalable, and can be used by farmers who are not very knowledgeable in the technical side. Concisely, this project will focus on improving agricultural output, minimizing crop wastage, and equip farmers with technology-based solutions that are smart and reliable in promoting sustainable agricultural practices.

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Table of Abbreviation

Abbreviation	Full Form
DL	Deep Learning
ANN	Artificial Neural Network
CV	Computer Vision
GPU	Graphics Processing Unit
CPU	Central Processing Unit
IoT	Internet of Things
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
XML	Extensible Markup Language
JS	JavaScript
UI/UX	User Interface / User Experience
DBMS	Database Management System
SQL	Structured Query Language
NoSQL	Not Only Structured Query Language
CRUD	Create, Read, Update, Delete
IDE	Integrated Development Environment
MVC	Model View Controller
SaaS	Software as a Service
IaaS	Infrastructure as a Service
PaaS	Platform as a Service
CDN	Content Delivery Network
DNS	Domain Name System
IP	Internet Protocol
OTP	One-Time Password
SSL	Secure Sockets Layer
TLS	Transport Layer Security
KPI	Key Performance Indicator
ETA	Estimated Time of Arrival
FAQ	Frequently Asked Questions

1. Introduction

1.1. Project Description

The object of work of this Final Year Project is the design and development of an AI-Powered Crop Disease Detection and Recommendation System which is planned to modernize the archaic agricultural activities by the means of intelligent automation. Many economies rely on agriculture as a major sector but the crop diseases are leading to a lot of losses in yield as a result of late diagnosis, misdiagnosis, and inability to get professional advice particularly in rural and remote regions.

The proposed project is going to solve these issues by offering farmers with a technology based method of offering early disease detection and disease treatment services. The proposed system allows farmers to take or post photos of the leaves of the crops using a simple web or mobile interface. These images are then subjected to computer vision algorithms and deep learning architectures or models namely Convolutional Neural Networks (CNNs). CNNs perform very well in image-based processes because they are able to automatically identify features that are important to the task including color change, spot pattern, texture irregularity and leaf deformity amongst others that suggests the existence of the disease. The model that is trained is used to match the uploaded image with a database of labeled crop disease images in order to categorize the disease correctly or healthy crops.

As soon as the disease is detected, the system creates real-time diagnostic outcomes and individualized treatment prescriptions. These can be recommendations based on proper application of pesticides, organic treatments, nutrient modification and preventive measures to prevent further spread of the disease. It focuses on sustainable agricultural practice by encouraging environment friendly solutions and reducing the massive use of chemicals thus ensuring the health of soil and the environment.

Besides diagnosis and recommendations, the system has the capability of storing the historical data in terms of disease occurrences, type of crops and the seasonal patterns. The data can be considered to find trends that would enable the farmers make data-driven decisions regarding future planting Project periods and disease management techniques. Agricultural researchers and policy makers can also find such insights useful in the patterns of diseases in regions.

Comprehensively, this project is evidence of how artificial intelligence can be applied in the farming industry, ensuring that farmers have a cheap, scaled, and accessible solution. The system is able to mitigate crop losses, increase productivity, and food security by incorporating deep learning, image analysis, and expert knowledge. The project eventually helps in the development

of smart and sustainable agriculture where farmers are able to access information that is timely, accurate and takes action.

1.2. Current Scenario

1.2.1. Local Scenario

Agriculture is a major economic activity in Nepal and other agricultural economies with a significant number of people living on it. Most farmers engage in small-scale or subsistence farming and they greatly depend on traditional knowledge and experience to take care of crop health. Identification of crop diseases in these areas relies on manual visual identification which many times cannot be adequate to identify any disease with similar symptoms. This has resulted in poor diagnostics thus more crops are lost through ineffective treatment.

Professional agricultural services, including those of plant pathologists, agronomists or government extension officers are inaccessible, especially in remote and rural areas. Farmers also tend to rely on informal recommendation by fellow farmers, local suppliers or community networks and they may not be accurate, or scientifically proven. Moreover, the time spent on consulting the specialists and transport limitations and financial constraints often lead to the inability to diagnose the disease in time and spread it quickly and lead to the consequences of its spread on crops.

The other major issue in the local environment is incompatible internet connectivity. The network coverage, bandwidth or unstable access to the digital services is the norm in many rural locations in Nepal. This limits the ability of farmers to utilize data intensive farming systems or real-time online consultancy. Thus, any technological solution that is supposed to work in this environment must be light, portable, and able to operate within low-connectivity conditions, e.g., by compressing images, means of caching the data offline or by synchronizing with the offline data.

Moreover, the digital literacy of farmers is rather different, which is why simplicity and ease of use should be one of the key design factors. Adoption can be greatly increased with a system that is easy to use, as it has clear and visual feedback and localized recommendations. Through these localized problems, an artificial intelligence-based crop disease detection system can provide accessible, timely, and practical assistance to Nepalese farmers and reduce their crop losses to improve their productivity and the sustainability of the agricultural industry despite infrastructural constraints.

1.2.2. Global Scenario

Crop diseases are one of the primary threats to agricultural productivity and food security at the global scale as they bring about significant losses in the yields of crops annually in the developed and developing countries. Pathogens such as fungi, bacteria, viruses, and pests which cause plant diseases lower the quality and quantity of crops and lead to losses to farmers and elevated food prices across the globe. The agricultural research organizations reveal that a good portion of the total crop output in the world is every year wasted on plant diseases that could have been prevented or detected late.

This problem has been increased by the effects of climate change. The increase in temperatures, alteration of rainfall patterns and the increase in humidity have provided the conducive environment with which crop diseases can spread very fast to a new geographical location. Previously, region-specific pathogens have been seen to attack crops in different continents, which complicate and predict the disease burden. The changing global environment requires more adaptive and quicker disease detection systems.

In spite of the fact the developed diagnostic methods like laboratory testing, hyperspectral imaging, and expert-based agricultural advisory systems are in use in some places, they are usually costly, infrastructure-based, and need special skills to work. This means that they can be used mostly in large-scale commercial farms or research institutions. Smallholder farmers who form a great percentage of the global agricultural labor force are not often able to access these advanced tools either because of cost, technical complexity and geographical location.

The recent years have seen the rise of AI-based image classification through deep learning model as a potential solution in the world to detect crop diseases. Such systems are able to scan photographs of plant leaves and diagnose accurately diseases in real time with devices that are readily available like smartphones. Their potential is not yet widely used by smallholder farmers as they face some barriers including low levels of digital literacy, insufficient access to the internet, absence of localized disease data and insufficient awareness/confidence in AI-based solutions.

Therefore, the global scenario highlights the need for affordable, accessible and farmer-centric AI solutions that can be adapted to diverse agricultural environments. Bridging the gap between technological innovation and practical adoption is essential to ensure that AI-powered disease detection systems deliver real-world impact at a global scale.

1.3. Problem Domain and Project as a Solution

1.3.1. Problem Domain

This project problem domain lies in the imperative issues of agricultural sector caused by crop diseases that are a significant threat to agricultural productivity, livelihoods of farmers, as well as food security in the world. Fungus, bacteria, viruses, and pests infecting crops are transmissible and can cause massive losses in yields and cost of production unless detected at the initial stage. The small-scale and subsistence farmers are especially susceptible because they generally cannot access prompt diagnostic equipment and the professional farm services.

Historically, farmers visually identify the disease through agricultural inspection or use of the services of agricultural experts. This method is time-consuming, subjective, and inaccurate in nature since the symptoms of disease may change with different types of crops, different stages of growth and the environmental conditions. Furthermore, most rural and isolated agricultural societies do not have access to professional agronomists, laboratories, or extension services thus, professional advice is expensive and unfeasible. This often causes farmers to use guesswork or late measures, which may increase disease outbreaks and cause too much or improper use of pesticides.

The domain of this problem is at the cross-point of agriculture, artificial intelligence, and information systems, where the need of automated and scalable disease detection solutions that are accurate and free of mistakes is increasing. Recent developments in the field of computer vision and deep learning, specifically, Convolutional Neural Networks (CNNs), allow impressive visual pattern analysis of images of crop leaves and the detection of diseases with high accuracy.

Overall, the problem domain highlights the urgent need for intelligent, data-driven agricultural solutions that reduce dependency on manual expertise, enable early disease detection, support sustainable farming practices and ultimately enhance food security and rural economic stability.

1.3.2. Project as a Solution

The suggested AI-Powered Crop Disease Detection and Recommendation System is a viable and scalable answer to the problems of the traditional process of crop disease diagnosis. The project will offer affordable diagnostic platform to help farmers identify crop diseases within a short time and accurately without involving experts all the time by combining artificial intelligence, computer vision, and information systems.

The system enables farmers to post images of leaves of crops using an easy to use web or mobile platform, meaning that the solution can be used even by people with minimal technical expertise. After an image is provided, it is fed into one of the trained Convolutional Neural Network (CNN) models, which has been trained to identify disease trends based on a vast collection of labeled image crops. This model examines the visual features like discoloration, texture, spots and lesions to determine the most probable disease to the crop. This automated system saves on a lot of time in diagnosing and the risk of human error is minimized.

Once the disease prediction is created, the platform gives customized treatment advice to help farmers act instantly and make an informed decision. These suggestions can involve organic solutions, chemical pesticide advice, manure and nutrient proposals and precautions to minimize future disease incidences. The system provides a variety of treatment options, which promotes sustainable and responsible farming systems, enabling the farmers to select methods that can best fit their resources and environmental requirements.

Besides the real-time diagnosing, the system will archive past prediction records such as type of crop, disease and the time the disease occurred. This information would also be processed to create visual analytics and reports on diseases trends, so farmers can understand what problems are recurring, what are the disease patterns during the seasons and how effective the treatments are. These insights can be used in the decision making that is data-driven and long-term crop management.

i. Core Solution Capabilities

- Disease detection through CNN based deep learning model.
- Output of disease within seconds following image uploads.
- Fertilizer, pesticide and organic treatment suggestions.
- Disease trends and frequency patterns visualized with the use of charts.
- All predictions have a history, which is guaranteed by cloud-based storage.
- Is a Progressive Web App (PWA) that can work on mobile and low-bandwidth environments?
- Admin dashboard to add, manage or update disease datasets.

The best thing about this system is that it is not only able to diagnose but also guide, monitor and educate. It also does not end at the identification farmers are made structured recommendations to treatment that are in line with sustainable agricultural practices. Past performance is also recorded in the system, which makes farmers to keep track of the performance and observe the patterns of recurrence over time.

ii. How This System Solves Existing Challenges

Table 1 : Table of Traditional Problem and AI-Based Solution

S.N	Traditional Problem	AI-Based Solution
i.	Slow and inaccurate manual identification.	Instant disease prediction powered by CNN.
ii.	Lack of expert access in rural areas.	Accessible from any device with internet.
iii.	Confusion between similar disease symptoms.	High accuracy pattern recognition model.
iv.	No documentation for case history.	Auto saved prediction logs and reports.
v.	Overuse of pesticides.	Balanced treatment and prevention suggestions.

iii. Direct Benefits to Farmers

- Faster diagnosis → early treatment → reduced crop loss
- Increased yield and revenue due to timely intervention
- Lower dependence on physical experts and middlemen
- Better planning for future farming cycles through history analytics
- Improved soil and crop health with informed chemical usage

Overall, this is a project that converts conventional farming to smart farming, which is a combination of technology and field viability. The AI-powered system can serve as a digital agricultural assistant through which with a touch of a button, farmers can obtain accurate, empowering and more direct and timely information, able to work round the clock, scale easily and enable farmers to make reliable decisions. This system has the potential to make agriculture efficient, intelligent and resilient with the ability to diversify its crops, types of diseases and health monitoring.

1.4. Aim and Objectives

The primary aim of this Final Year Project is to create a full-fledged end-to-end AI-based crop disease detection and advisory system that will be able to interpret the images of plant leaves correctly and provide a diagnosis and a treatment prescription in real time. The system aims at assisting the agricultural sector by decreasing reliance on the manual diagnosis, reducing the financial loss as a result of disease outbreaks, and facilitating the farmer in making better farming decisions with the help of the digital solution. This project will fill the gap between the early

detection of diseases and taking corrective measures on time by incorporating Artificial Intelligence, deep learning, REST API communication, and dashboard analytics based on MERN.

In its essence, the project is aimed at enhancing agricultural productivity, sustainability and accuracy of decisions. The traditional agriculture mostly relies on experience-based recognition that is liable to misunderstanding and slowness. Compared to it, this AI system facilitates the prediction of the disease through visual identification by comparison with a learned set of symptoms so that even unprofessional users can identify the problem in a short period of time and with high accuracy. With the additions of history tracking, notification, and visual analytics, this project is not only a detection platform, but an entire agricultural intelligence platform.

1.4.1. Aim

The main aim of the project is to create and innovate an AI-powered web application that can detect the disease of crops based on uploaded photos of leaves, give correct treatment information and give analytical data to help farmers avoid crop loss and enhance agricultural production.

1.4.2. Objectives

The objectives of the project are given below:

i. Technical Development Objectives

- To train a CNN-based deep learning model using an agricultural dataset for disease classification.
- To implement a Flask API microservice that processes uploaded images and returns prediction results.
- To develop a MERN-based web interface where users can upload images and view results.
- To store prediction logs, user records, and disease data in a scalable MongoDB database.
- To integrate analytical graphs using Chart.js/Recharts for trend visualization.
- To develop a Progressive Web Application (PWA)-based UI for mobile accessibility.
- To support history tracking, notification alerts, and offline caching for usability.

ii. Functional Output Objectives

- To automatically detect diseases with high accuracy and confidence percentage.
- To provide remedy solutions including:
 - Organic remedies
 - Fungicides & pesticides
 - Fertilizer recommendation & dosage

- To store past predictions for monitoring disease patterns over time.
- To allow admin users to update disease categories, retrain the AI model & manage users.
- To generate region-wise disease frequency reports and usage stats for agricultural study.

iii. End-User Benefit Objectives

- Minimize crop damage by diagnosing diseases at early stages.
- Reduce farmers' dependency on external consultancy for basic diagnosis.
- Increase crop production through fast intervention and guided treatment.
- Improve resource management by reducing wasted pesticide usage.
- Enable farmers to digitally track plant health progression across seasons.

1.4.3. Why This Objective Matters

With the achievement of these goals, this project will directly affect the smart agriculture transformation which will enable technology to substitute the uninformed decision-making process with the data analysis. Farmers are not obliged to wait days waiting the inspection of the experts but they get the results immediately and make sure that the treatment is provided at the appropriate time, thus avoiding the spread of diseases and loss of money. This system can be scaled, be future ready and able to move to other types of crops and monitor agriculture at a national level, thanks to the combination of AI and dashboard analytics.

1.5. Structure of the Report

1.5.1. Background

The background section of this report gives a detailed insight into the context, in which the AI-Powered Crop Disease Detection and Recommendation System is developed. Agriculture is a crucial aspect of the global economy, more so among the developing nations such as Nepal where a significant percentage of the population relies on farming as a means of sustenance. Nevertheless, crop illnesses have continued to be one of the greatest problems that have impacted agricultural output and food security across the globe (Strange and Scott, 2005). Conventional disease detection techniques are based on manual inspection and/or consultation with agricultural experts, which have been found to be quite time consuming, subjective and inaccurate in certain cases (Bock et al., 2010).

The project is designed to overcome these shortcomings by incorporating the latest technologies, including Artificial Intelligence (AI), computer vision, and web-based systems. Convolutional

Neural Networks (CNNs) have proven to be highly successful in AI-based image classification, in identifying plant diseases with high accuracy (LeCun et al., 2015). The background also comprises a background on the target users of the system. Small-scale farmers are the major users as they require a low-cost and user-friendly solution to detect crop diseases. Secondary users are agricultural experts and system administrators, who deal with system accuracy and reliability.

Moreover, the solutions like PlantVillage and LeafSnap prove the efficiency of AI use in plant diseases recognition, but they do not provide local recommendations and are not useful to farmers (Mohanty et al., 2016). Commercial systems such as Cropix are very advanced but costly and difficult to use by small scale users.

On the whole, this section indicates the necessity of an accessible, practical, and smart system that would fill the gap between the high-tech AI applications and the actual agricultural needs.

1.5.2. Development

The development section of this report dwells upon the methodologies and design principles and implementation strategies according to which the AI-based crop disease detection system is built. Since AI-based projects are dynamic, with requirements changing according to the model performance and user feedback, a flexible and iterative development process is necessary. Having considered multiple approaches, like Waterfall, Agile, Rapid Application Development (RAD), and Prototyping, the Scrum model under the Agile approach was chosen as the most appropriate (Sommerville, 2016).

Scrum breaks up a project into smaller iterations (sprints) that enable on-going development, testing and refinement. Such an iterative method is especially applicable to the AI systems in which the accuracy of the model and system performance should be continually refined (Schwaber and Sutherland, 2020). The process involved requirement gathering, system design, implementation and testing.

Technologically, the system has been created based on the MERN stack, comprising of MongoDB, Express.js, React.js and Node.js. This stack enables scalable and efficient full-stack web application development (Flanagan, 2020). A Convolutional Neural Network (CNN) is the key element of the disease detection functionality: this model is popular in image recognition tasks because it can identify complex visual features (LeCun et al., 2015).

System design consists of diagrams like Use Case Diagrams, Data Flow Diagrams (DFD) and Entity-Relationship Diagrams (ERD) that aid in illustrating the structure of the system and data

flow. RESTful APIs enabled communication between system elements and the data was exchanged efficiently (Fielding, 2000).

Overall, the development process emphasizes flexibility, scalability, and user-centered design, ensuring the system is both technically robust and practically usable.

1.5.3. Testing and Analysis

The testing and analysis section is a way to measure the performance, reliability and effectiveness of the developed system. Software testing is an important step in software development because it helps to verify that the system works as anticipated and fulfills user requirements (Pressman, 2014). For this project, both unit testing and system testing approaches were used.

The individual components that were unit tested included the functionality of login, uploading of images, AI prediction, and generation of recommendations. All modules were tested separately to test their correctness and stability. The system testing was done to test how all the components worked together and how well the system worked.

The performance of the AI model was studied in terms of the accuracy of predictions and confidence. It has been demonstrated that CNN-based models can be effectively used in image classification tasks, especially in the agricultural industry, where they can detect disease patterns with high accuracy (Mohanty et al., 2016). The system was shown to be fairly accurate in identifying crop diseases, but the quality of the training data influences the performance.

There was also analysis of the survey to assess user satisfaction. The outcome of pre-survey indicated that farmers had difficulties with disease detection and post-survey results indicated enhanced understanding and system satisfaction.

Nevertheless, certain limitations were also found such as reliance on internet connection and less accuracy when dealing with unknown diseases. Nonetheless, the system can greatly enhance the efficiency of detecting the disease and make more effective decisions irrespective of these issues.

1.5.4. Conclusion

The conclusion section will summarize the main findings and contributions of the project. This AI-Powered Crop Disease Detection and Recommendation System reflects the application of artificial intelligence in agriculture and its potential to solve the problems in practice. The system would assist in minimizing losses to crops as well as enhancing agricultural productivity, since it

facilitates the detection of diseases at an early stage, and it offers a recommendation on how to act (Strange and Scott, 2005).

A user-friendly design is another key strength of the system in that it enables farmers with little technical expertise to operate the system satisfactorily. Combining AI-images recognition with real-world treatment recommendations will make sure that the system offers the diagnosis and solutions. Moreover, data storage, analytics, among other features, can be used to support long-term planning and decision-making.

Although there are benefits, the system has limitations. Predictability is based on the data set to be trained on the AI model, and the system needs to be connected to the internet to process in real time. These constraints demonstrate that additional enhancements must be made, including increasing datasets and providing offline capabilities.

Further work can involve the addition of support to more crops, better model accuracy, and the incorporation of IoT-based monitoring systems. The system can be scaled and deployed on a large scale with its further development.

To sum up, the project shows the potential of AI as a transformer in agriculture, and it is an excellent premise to smart farming solutions of the future (LeCun et al., 2015).

2. Background

2.1. About the End Users

The main consumers of the proposed AI-based crop disease detection system will be farmers and specifically small-scale farmers residing in rural and semi-urban areas. Such farmers are usually highly dependent on agriculture as their primary income and directly influenced by the crop diseases, climatic variations, and the lack of access to the expert assistance in a timely manner. Smallholder farmers do not have high-level technical expertise or access to agricultural extension services on a regular basis, and therefore, early detection and correct treatment advice is a serious problem. The system is thus expected to be straightforward, cheap yet accessible so that farmers can post crop leaves images via a mobile phone or web interface and get instant disease forecasting and treatment prescriptions.

Other secondary users such as agricultural experts and system administrators are also needed in the system as they are instrumental in ensuring that the platform remains accurate and reliable. Agricultural specialists will ensure that the classification of the diseases, the recommendations of treatment options, and the knowledge base are updated according to the emergence of crop diseases, seasonality and agricultural practices in different regions. Their participation will see to it that the system offers scientifically valid and context-specific advice to farmers.

The administrators of the system will take care of overall management and monitoring of the platform. They have the responsibility of maintaining, managing user accounts, monitoring system performance, providing data security, as well as updating systems. The administrators are also able to look into data and disease trends generated by the system in aggregation and can be used to facilitate decision-making, policy formulation, and even future upgrades to the platform. All these groups of users make the system accurate, scalable, and sustainable agricultural development beneficial.

2.2. Understanding the Solution

The solution offered by the author is implemented in the form of Artificial Intelligence (AI), the newest web technologies, and cloud computing to provide a smart and conveniently designed crop disease detection system. The system will assist farmers in early detection of crop diseases and timely treatment advice on them, hence minimize loss of crops and enhance agricultural yield.

The workflow of the system starts with the acquisition of images where the users will be allowed to upload pictures of crop leaves via web or mobile application. These images are subjected to an image preprocessing step, which involves image resizing, image normalization and image noise reduction to maintain uniformity and enhance model accuracy. The preprocessed images are sent to the AI inference part where a trained Convolutional Neural Network (CNN) identifies visual patterns and labels the disease or verifies healthy crops.

After generating the prediction, the system goes to the result generation stage where the identified disease name, the level of confidence, information on the severity is assembled. The system provides custom-made recommendations based on the diagnosed illness, including organic treatment recommendations, chemical recommendations, fertilizer recommendations, and preventive recommendations. Agricultural experts curate and validate these recommendations in order to guarantee the reliability and the relevance of the recommendations to the region.

The solution also gives great consideration to usability, which makes the interface easy and not hard to use by people with minimal technical knowledge. It is accurate with the help of ongoing training of the models and changing of the data and is scalable with the well-known cloud-based infrastructure that enables the system to operate with the increasing number of users and data satisfactorily. Moreover, the system will also contain Progressive Web App (PWA), whereby it can store the results and recommendations that have already been seen by the user, and they can use them even without internet access, which is especially useful in those locations where the net connection is weak or unreliable.

In general, the concerted solution guarantees the creation of a solid, scalable, and user-friendly solution that will close the distance between the high level of AI technologies and the actual demand of agriculture

2.3. Similar Projects

2.3.1. Project 1 – PlantVillage

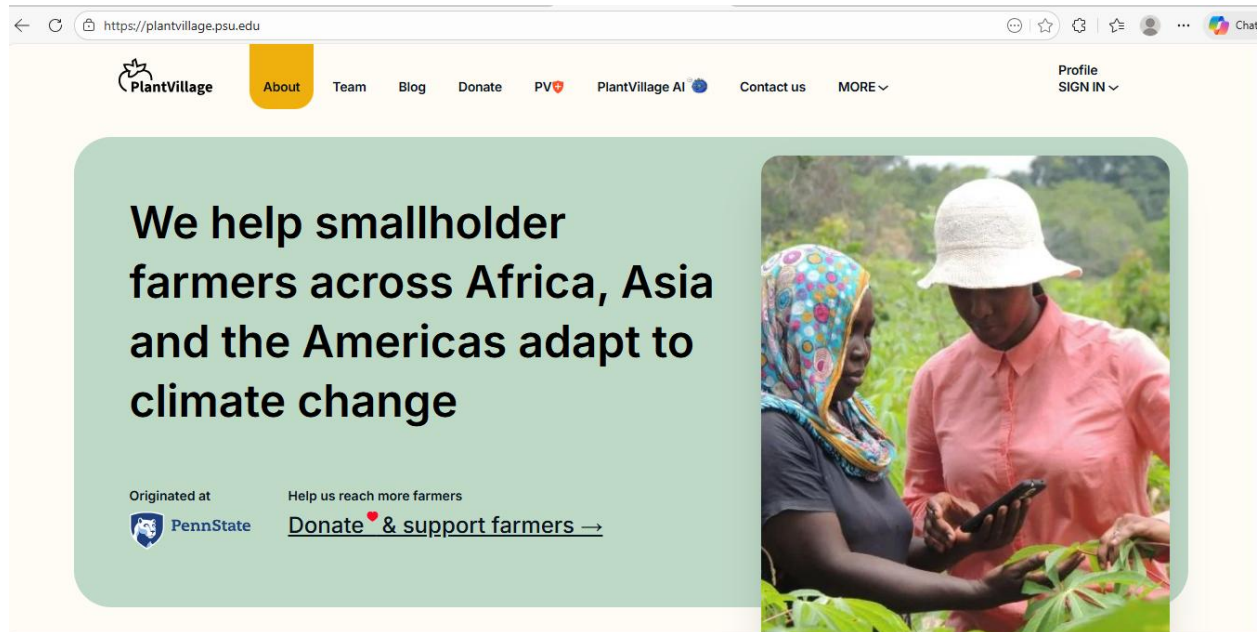


Figure 1: Similar Project number 1

Link: plantvillage.psu.edu

PlantVillage PlantVillage is a globally known open-source dataset and deep learning-powered platform created to assist in identifying plant diseases with the help of images. It offers a big and well organised set of labelled pictures of plant leaves that have been widely used to educate and assess Convolutional Neural Network (CNN) models. PlantVillage is highly successful in detecting a variety of crop diseases because of the quality and volume of its data, and it has been an important part of agricultural AI research.

Although the performance of the PlantVillage is good in disease classification, it is rather oriented on disease identification and not end-user support. The site provides a few localized treatment guidelines, and it fails to sufficiently take into account area-specific agriculture, weather, or resource supply. Also, the system is less farmer-focused and more research-driven, not having such features as simpler user interfaces, offline capabilities, support in multiple languages, and monitoring disease history.

Consequently, although the idea of PlantVillage is an ideal starting point to train models and benchmark, it does not take the practical requirements of smallholder farmers into consideration to the full extent. This drawback shows the need of a more holistic solution that would include proper detection of the disease along with localized guidance and actionable and usability effects, which this proposed system would offer.

2.3.2. Project 2 – LeafSnap

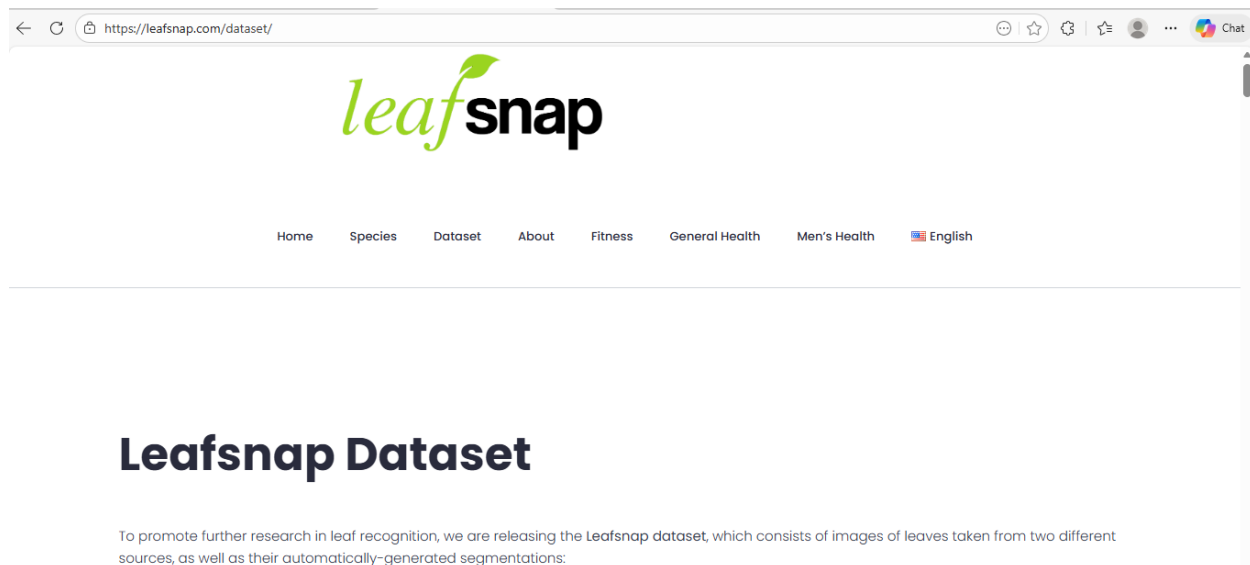


Figure 2: Similar Project number 2

Link: [Leafsnap Dataset - LeafSnap](https://leafsnap.com/dataset/)

LeafSnap is an application created by the company depending on image recognition and machine learning algorithms to identify plants by their leaves with the use of an image. The system matches visual data of plant leaves like shape, texture, and patterns of veins in a pre-existing database and uses the information to classify the plant species. LeafSnap has become popular in educational, botanical and environmental applications.

Nonetheless, the main task of LeafSnap is not the diagnosis of crop diseases, but the classification of plants. The application does not interpret symptoms associated with the health of plants or offer information on the causes, severity, and treatments of the diseases. It also does not have agricultural decision support capabilities as far as preventive advice, fertilizer advice or currently validated farmer solutions.

Therefore, though LeafSnap shows that image-based plant recognition works, there are no issues regarding how farmers can practice the idea of managing crop diseases in practice. This difference also serves as the reminder of the necessity of a special system aimed at detecting the disease, suggesting treatment, and providing support to the farmers, which is the idea of the given project.

2.3.3. Project 3 – AgriDoctor

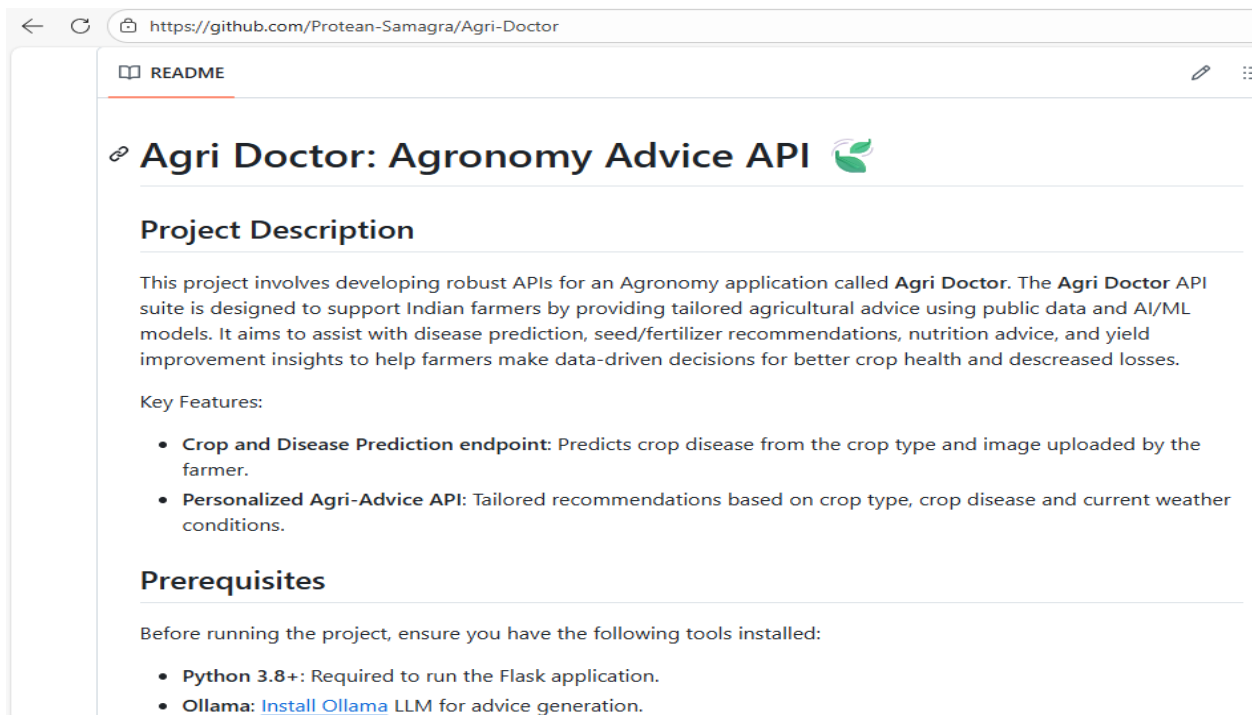


Figure 3: Similar Project number 3

Link: [Protean-Samagra/Agri-Doctor](https://github.com/Protean-Samagra/Agri-Doctor)

AgriDoctor is an agricultural decision support system which gives information about crop disease via expert system technique. The interface depends on logic-based rules and human expertise, where the user manually enters the symptoms he/she sees (e.g., the color of the leaves, spots, patterns of growth). According to these inputs, the system compares the symptoms with a set of rules stored to detect potential diseases and propose general management principles.

Although AgriDoctor can be handy in the delivery of expert-validated knowledge, it remains very susceptible to error in case of manual input that is given by people. The farmers will not be able to identify or describe symptoms appropriately thus making incorrect or incomplete diagnoses. Also, the system is not very adaptable to new or changing diseases and it is hard to scale unless expert intervention continues, as it is rule-based.

Moreover, AgriDoctor does not have an AI-based image analysis, automatic learning, and performance optimization in real-time. Consequently, even though it can be a useful source of expertise, it is not as automated, accurate, and user-friendly as it needs to be to be adopted by smallholder farmers in large scale. These shortcomings indicate the benefits of the suggested system, combining AI-powered image identification and professional verification of the recommendations to provide a more convenient and efficient solution.

2.3.4. Project 4 – Cropix

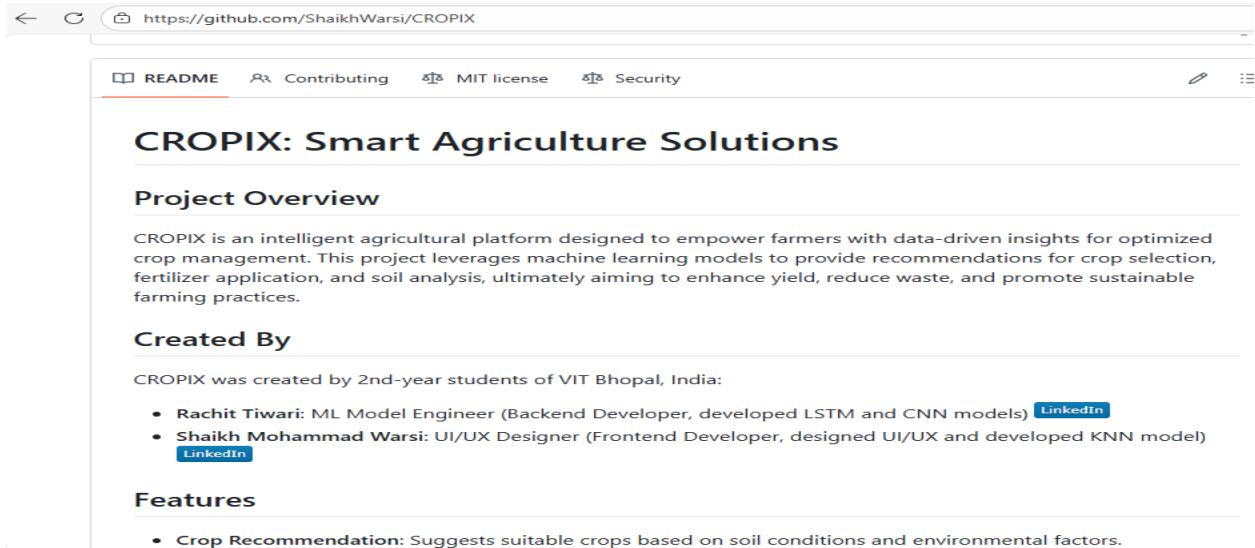


Figure 4: Similar Project number 4

Link: [ShaikhWarsi/Cropix: An AI-driven solution that uses data insights to help farmers increase their productivity and efficiency.](https://github.com/ShaiKhWarsi/CROPIX)

Cropix is a commercial agricultural technology platform offering AI-powered crop disease detection and advanced analytics and farm monitoring technology. The system has the ability to detect plant diseases based on the analysis of photographs and also provides other services like crop health, yield information and data-informed decision support. Cropix is very acceptable in large scale farms and agribusinesses because of its advanced analytical functions.

Cropix is also expensive, though it has technical advantages, and may be necessary to include subscription or special hardware or integration on an enterprise level. This renders the platform not accessible to the smallholder farmers, especially in the rural and developing areas where financial and technological capabilities are scarce. Moreover, the system can end up being too complex to effectively use by farmers who are not very digital literate.

Consequently, although Cropix can show the promise of high-tech AI and analytics in the agricultural sector, it fails to meet small-scale farmers in the context of their affordability and accessibility requirements. This constraint supports the necessity of such a solution that is cheap, user-friendly, and publicly available, which the offered solution should offer by integrating AI-driven disease detection with features such as farmer-focused design and offline support.

2.3.5. Project 5 – FarmAI Research Systems

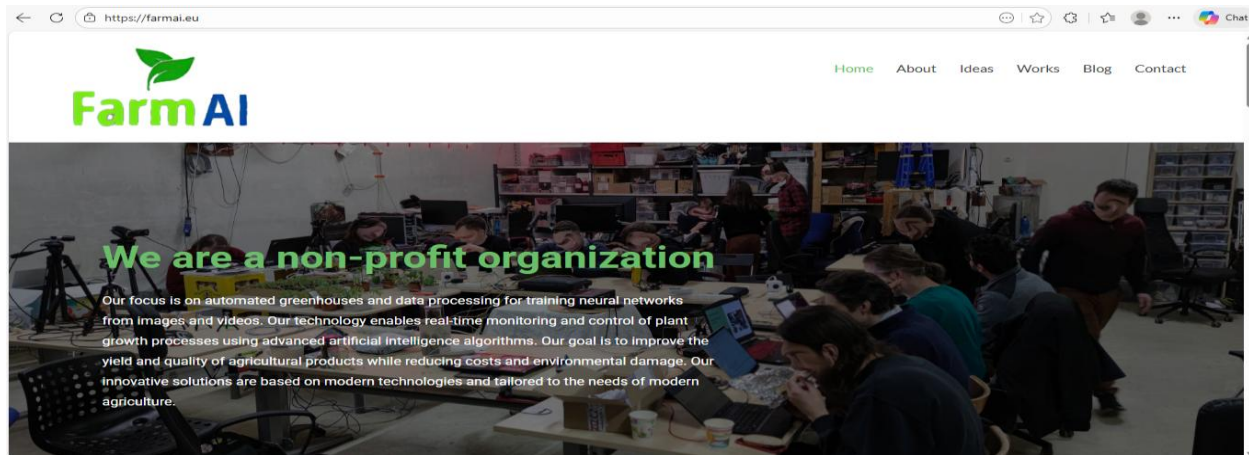


Figure 5: Similar Project number 5

Link: [FarmAI – Neural network training, automated greenhouses](https://farmai.eu)

FarmAI research systems are a body of scholarly and practical AI-based applications that have been proposed to identify crop diseases with state-of-the-art machine learning and deep learning algorithms. Such systems are frequently highly classified on controlled environments and are a source of innovation and research on agricultural technology. They are typically tested in comparison to benchmark datasets and are used to investigate new algorithms, architectures and optimization methods.

Nevertheless, the majority of the FarmAI research systems are not ready to be deployed and do not consider practical issues to apply in practice. They are usually high-computational processes, regulated image input, and laboratory-based processes and thus impracticable to direct farmers. Moreover, these systems are typically not user friendly, not local in their recommendations, offline, and long term strategy of maintaining the system.

Therefore, although FarmAI research systems show the technical possibility and potential of AI in crop disease identification, they do not bridge the research and the application in the real-life agricultural contexts. This constraint makes the need to come up with a solution that is both highly accurate and at the same time focuses on usability, accessibility, scalability and real-world implementation as the primary goals of the proposed system.

2.4. Comparisons

A thorough analysis of the current crop disease detection and agricultural decision-support systems reveals that even though there has been considerable advancement in employing artificial intelligence and expert knowledge in agriculture, most solutions are applied to a single component of the issue. Consequently, it is evident that there is still a significant disparity between the technically developed systems and solutions that are actually applicable and available to the farmers in the actual real world situations.

PlantVillage and other research systems of FarmAI have shown that, with properly trained deep learning models, there is a high potential of predicting diseases in crops with high accuracy using leaf images. These systems are also useful tools in the academic research and have served to set benchmark datasets and evaluation standards. They are however, mostly geared into controlled conditions and testing. They are usually not deployment-ready, including user-friendly interfaces, localized treatment advice, multilingual supports, and offline accessibility. These restrictions are very limiting to the practical usefulness of smallholder farmers, particularly those who reside in rural regions with little connectivity and technical literacy.

Applications such as LeafSnap also portray the issue of scope limitation. LeafSnap is a successful project with its main aim being classification of plant species though it takes advantage of image recognition to identify the species of plant instead of crop health management. It does not examine the signs of diseases or offer treatment information, which is not appropriate to farmers who require actionable information to save their crops. That is why it is necessary to shift toward identification to the systems that are oriented on decisions and that directly assist the farming activities.

Other solutions based on expert systems like agri doctor do not follow this method but use pre-set rules and the knowledge of the expert. Although these systems have the potential to provide reliable information in case the inputs are precise, it relies on farmers who manually describe the symptoms. In reality, this is sometimes challenging, since farmers might fail to recognize the signs of a disease, or they might lack the technical terminology to do so. Also, systems that are based on rules are weak in adapting to new diseases and climate changes and to receive new agricultural conditions, which limits their scalability and long-term performance.

The most sophisticated side of the spectrum is commercial platforms such as Cropix which are a combination of AI-based disease detection, analytics and monitoring. Nevertheless, small-scale farmers are often unable to have them because they are very expensive, they are subscribed to and are also complicated to operate. These types of solutions are normally laid down in large farms or agribusinesses, with a large proportion of farming community poorly served.

Comparatively, the suggested system is to be set to balance between accuracy, affordability, usability and applicability to real-life. It combines AI-driven image detection with expert-approved, localized treatment suggestions based on local farming methods. The simplicity of the interface along with intuition is to guarantee easy use by the farmers with little technical skills. Moreover, the scalability is facilitated by the use of cloud services, and Progressive Web App (PWA) characteristics allow to access the results already seen offline and solve the issue of connectivity that is typical of rural environments.

The proposed solution is focused on the adoption of farmers and practical application, unlike research-only or high-cost commercial systems. It will fill the gap between technological advancement and the practical necessity of agriculture by focusing on the weaknesses of current projects and providing a cost-effective, inclusive, and impactful solution to the modern problems of crop diseases.

The following table represents a comparison between existing systems and the proposed AI-powered crop disease detection system based on key features and performance factor.

Table 2: The Table of comparison between existing system and the AI- powered system

Criteria	PlantVillage	LeafSnap	AgriDoctor	Cropix	Proposed System
Purpose	Disease dataset and detection	Plant identification	Rule-based diagnosis	Commercial crop monitoring	AI-based disease detection & recommendation
Disease Detection	Yes (image-based)	Limited	Yes (rule-based)	Yes	Yes (AI-powered CNN)
Accuracy	High (dataset dependent)	Moderate	Low–Moderate	High	High (improves with training data)
Treatment Recommendation	Limited	No	Yes	Yes	Yes (detailed & practical)
User Interface	Moderate	Simple	Basic	Advanced but complex	Simple and user-friendly
Target Users	Researchers, developers	General users	Farmers	Large-scale farmers	Small & medium-scale farmers
Cost	Free	Free	Free	Paid	Low-cost / Affordable
Internet Requirement	Yes	Yes	No	Yes	Yes (future offline possible)
Localization (Region-specific advice)	No	No	Limited	Yes	Yes
Ease of Use	Moderate	Easy	Easy	Moderate	Very easy

Criteria	PlantVillage	LeafSnap	AgriDoctor	Cropix	Proposed System
Technology Used	Deep learning datasets	Image recognition	Rule-based system	AI + IoT	AI (CNN) + MERN stack
Scalability	High	Moderate	Low	High	High
Limitations	Not user-focused	No disease advice	Less accurate	Expensive	Dataset dependent

The proposed system addresses these shortcomings by integrating precise AI-driven disease detection with actionable recommendations, cost-effectiveness, and usability, which makes it more applicable to the practical application in agriculture.

3. Development

3.1. Considered Methodologies

One of the most important decisions in the life cycle of a software project is the choice of a development methodology because it directly determines the effectiveness, quality, and success of the system under development as well. In our situation, the project is the creation of an AI-based system of crop disease detection, which includes such sophisticated elements as the models of artificial intelligence, image-processing algorithms, and the web or mobile interface. The combination of these elements contributes to further complications of the project and, thus, the selection of the methodology becomes even more significant. There are seldom defined or set requirements in AI-based projects unlike in a traditional system. Machine learning models change in accuracy, data quality and variety, and end-user expectations can vary considerably during development. As an example, pre-testing of the disease detection model can indicate the gaps in prediction accuracy or extra image preprocessing stages. Likewise, user feedback on the interface would necessitate modifications in the display of the system in terms of showing predictions and recommendations.

Due to such inherent variability, the methodology selected should be able to sustain flexibility, ongoing development, and improvement. It must enable the development team to try out alternatives of AI models, test algorithms, and take into account users without causing serious delays or rework. Meanwhile, the methodology must provide that the development is structured enough to be properly documented, tested and reliable of the system. Such two requirements of flexibility and organization are paramount to research and deployment-intensive projects such as ours.

We critically analysed various proven software development methodologies each of which are recognised such as the Waterfall model, Agile methodology, Scrum framework, Rapid Application Development (RAD) and Prototyping model to come up with the most appropriate methodology. All methodologies were evaluated on how they could cope with changing requirements, respond to creating iterative developments, encourage collaboration among stakeholders, and deliver functional increments in time. In doing this comparative analysis, we hoped to choose a methodology that would not only suit the experimental feeling of AI development but also give a clear outline of project planning, monitoring, and evaluation.

Finally, we concentrated on seeking a solution that would be flexible and controllable so that the system can be developed effectively and have high quality, usability, and reliability to the end-users.

3.1.1. Waterfall Model

The Waterfall model is an old fashioned model of software development that is linear and sequential in nature. The phased approach of requirements analysis, system design, implementation, testing, deployment, and maintenance should not be neglected and should be completed before moving to the next step. This model is suitable in cases of projects whose requirements are stable and well known (Pressman, 2014).

With AI-based systems, however, the requirements tend to evolve during the training and evaluation of the models. As an illustration, the accuracy of disease classification can be increased, the datasets can be enlarged, or other features can be ordered once preliminary testing is done. Waterfall model is rigid and thus, it is hard to accommodate such changes without major rework. Consequently, the Waterfall model was not deemed appropriate to this project because it lacked flexibility and did not support the aspect of iterative improvement.

3.1.2. Agile Methodology

Agile methodology is grounded on the principle of iterative and incremental development, according to which the system is developed in small and manageable steps with a continuous user feedback. Agile focuses more on flexibility, teamwork, and regular release of working software instead of large-scale advance documentation (Sommerville, 2016).

In the case of AI-based systems, Agile is especially useful, as it enables the training, testing, and refining of models in several cycles. User feedback and system reviews can be integrated into the next round of development within a short period of time. Agile also minimizes the chances of project failure since the problems are identified early and the system is also developed based on actual user requirements. Agile emerged as one of the good candidate methodologies due to these benefits.

3.1.3. Scrum Framework

Scrum is an orderly structure of Agile that breaks the development into short cycles, called sprints, which can take two or four weeks. Every sprint is concerned with delivering a workable and possibly deliverable increment of the system. Scrum comes with specific roles like Product Owner, Scrum Master and Development Team, which enhance accountability and communication (Schwaber and Sutherland, 2020).

Scrum will particularly be appropriate in this project since it facilitates the development of the AI in bits, including uploading images and disease prediction algorithms and recommendation systems. Frequent sprint reviews and retrospectives allow checking the progress and risks continuously, and Scrum is one of the methods that can be used to address uncertainty that can arise in AI experimentation.

Development of software and systems at a high pace (RAD)

3.1.4. Rapid Application Development (RAD)

RAD is concentrated on the quick development by the means of the rapid prototyping and minimum planning. Although the method is fast at the initial development stages, it usually compromises system architecture, documentation, and scalability (Pressman, 2014). Rad alone is not enough in case of an academic and research-oriented project which needs to be structured, analyzed, and maintained. So, it was not chosen as the main methodology.

3.1.5. Prototyping Model

The Prototyping model entails developing prototypes of the systems to ensure that the requirements of the users are clearly known and to obtain feedback. The method can be useful in cases where the requirements are not defined initially. Nevertheless, overdependence on prototypes may slow down the final development and lead to inadequately organized systems in case prototypes are not redesigned but constantly modified (Sommerville, 2016).

3.2. Selected Methodology

After carefully analyzing all the considered methodologies, the Scrum Framework under the Agile methodology was selected as the most suitable approach for the AI-based Crop Disease Detection System. This choice was driven by the project's iterative nature, the need for adaptability, and the structured management approach that Scrum provides. The development of this system involves multiple complex components, including AI model experimentation, software integration, and user interface development. Each of these components requires continuous testing, validation, and feedback, which Scrum accommodates exceptionally well through its sprint-based iterative cycles. Unlike traditional linear methodologies, Scrum allows flexibility to adjust requirements and refine

deliverables based on ongoing evaluation and stakeholder input, which is essential when working with AI models that may need repeated tuning for optimal accuracy.

Scrum organizes the project into short, time-boxed iterations called sprints, each lasting a few weeks and focusing on delivering specific functionalities. For this project, sprints are planned around critical milestones such as dataset preparation, AI model training, frontend and backend development, system integration, and testing. Breaking the work into manageable increments ensures that progress is measurable and any potential issues are identified early, preventing delays or major setbacks. Daily standup meetings provide a platform for immediate communication among team members, facilitating quick resolution of obstacles and enhancing collaboration. At the end of each sprint, sprint reviews and retrospectives are conducted to evaluate completed work, gather feedback, and plan improvements for the next sprint. This continuous review process allows for refinement of both the AI models and the application interface based on performance results and user feedback, ensuring that the system evolves toward optimal functionality.

The adoption of Scrum provides several key benefits for this project. Firstly, flexibility allows the team to adapt to evolving requirements, unforeseen challenges, and iterative improvements in AI model accuracy. Secondly, incremental delivery ensures that functional components are available for testing and evaluation at the end of each sprint, providing tangible results and early validation. Thirdly, Scrum encourages collaboration and communication among all stakeholders, which is critical for aligning AI performance, software development, and user interface design. Additionally, Scrum supports risk management by identifying and addressing issues early, thereby reducing the likelihood of major failures during the project. Finally, the methodology facilitates user feedback integration, allowing potential users, advisors, or stakeholders to test early versions of the system, ensuring that the final product meets real-world needs effectively.

By applying the Scrum Framework, the development of the AI-based Crop Disease Detection System is structured, efficient, and adaptable. It ensures that the project progresses in a coordinated and organized manner while maintaining high quality and timely delivery. The iterative approach enables continuous improvement, better integration of AI and software components, and a user-centered design, making Scrum the most appropriate methodology for this innovative and technically challenging project.

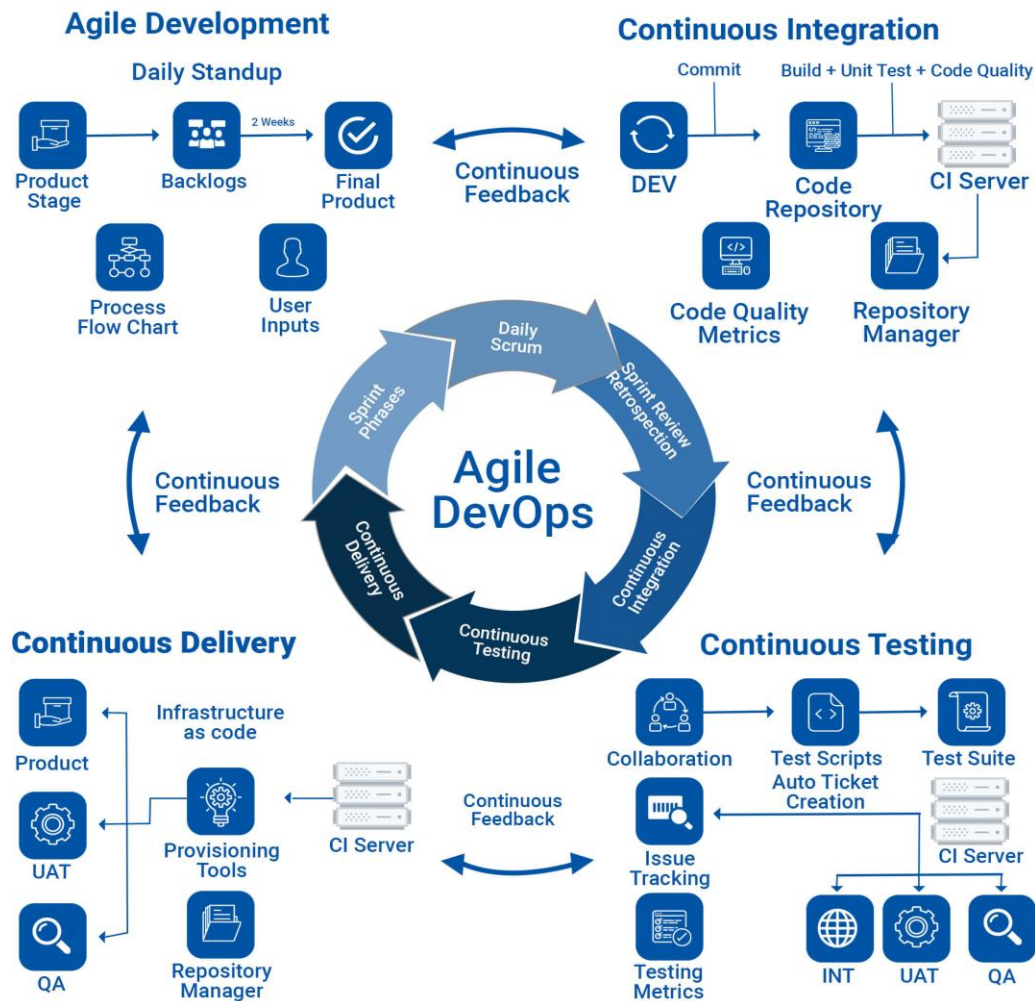


Figure 6: Selected Methodology

At last what we can say is that we chose the Scrum framework in this project after we have carefully considered the advantages and disadvantages of all the methodologies that were taken into consideration. Scrum offers flexibility necessary in the training of AI models, performance optimisation, and system requirement development. The sprint based framework will enable us to build upon and test the critical parts of the system, including precisely detecting diseases, prescribing treatment, and visualisation dashboards, in a step-by-step fashion.

The frequent review of sprints provides the focus on the project goals, whereas retrospectives promote the constant improvement of the process. Moreover, Scrum fosters the reduction of risks as it allows detecting technical problems and user concerns at the beginning of the development.

In general, Scrum is a structured, flexible, and feedback-driven approach that is the most suitable methodology in the successful development of this AI-driven system.

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3.3. Phases of Methodology

The design of the AI-Powered Crop Disease Detection and Recommendation System was an iterative and flexible process, relying on the Scrum framework, a commonly used Agile methodology used to organize complex software development projects (Schwaber and Sutherland, 2020). Agile techniques are especially applicable to projects based on artificial intelligence since they enable responding to the needs of the project in a constant way, frequent testing, and adapting based on the user feedback and model behavior (Sommerville, 2016).

The whole development life cycle was subdivided into four key stages: Planning, Development, Testing, and Review. The phases were linked and provided in an iterative manner to facilitate constant improvement and alignment to project goals.

3.3.1. Planning

The planning phase was the basis of the project and all the preparatory activities were done there, to establish the direction and scope of the system. This step included the identification of the

problem area, learning about the problems of farmers and examining what the current systems could not do. An elaborate need analysis was done by pre-surveys and informal interviews with the possible users in order to obtain a real world requirement and expectations.

The identification of functional requirements and non-functional requirements occurred in this phase. The functional requirements were features of image upload, disease detection and generation of the recommendations whereas the non-functional requirements were centered on the system performance, usability, scalability and security.

Various software development models like Waterfall, Agile and Rapid Application Development (RAD) were critically reviewed. Although the Waterfall model provides a well-organized structure, it does not have flexibility, and it would be inappropriate in the case of AI-based systems with frequent changes. The choice of the agile approach, especially Scrum, was made due to its ability to facilitate iterative development, constant feedback, and accelerated delivery of functioning elements (Sommerville, 2016).

A project plan was prepared in detail and it had task distribution, estimation of time and management of resources. Project timelines were represented with the help of such tools as Gantt charts, whereas Work Breakdown Structure (WBS) was applied to subdivide the project into manageable tasks (Kerzner, 2017). Risk analysis was also conducted to determine the possible difficulties like data constraints, model accuracy, and technical constraints.

3.3.2. Development

The development stage involved the conversion of the requirements planned into a functioning system through a methodical design and implementation. This stage was initiated by system design and different types of diagrams, including Use Case Diagrams, Data Flow Diagrams (DFD) and Entity-Relationship Diagrams (ERD) were developed to illustrate system functionality, data flow, and database structure (Pressman, 2014).

The system architecture was made such that it could be scaled, be modular and also allow effective communication between components. The MERN stack (MongoDB, Express.js, React.js, Node.js) was selected to develop, as it is flexible and can efficiently serve full-stack web applications (Flanagan, 2020). The frontend interface was created with React.js, providing a user-friendly and interactive interface, whereas the backend logic and API integration were implemented using Node.js and Express.js. The user data and prediction history and system records were stored in MongoDB.

The AI model is the main part of the system that classifies images with a Convolutional Neural Network (CNN). The computer vision application is one of the most common applications of CNNs that identify patterns in images (LeCun et al., 2015). The dataset on which the model was trained was a collection of crop leaf images, and thus this model was able to learn and discern patterns of diseases.

The development was also conducted in several sprints and each sprint was dedicated to the implementation of certain features like user authentication, image upload, prediction module, and recommendation system. The process of continuous integration and testing was carried out to maintain stability and performance of the system. The results of the early testing stages were used to enhance systems design and usability.

3.3.3. Testing

The testing stage was carried out to ensure and confirm that the system is within the desired requirements and that it works well under various circumstances. The software development lifecycle requires testing because it will assist in the process of detecting errors, enhancing the quality of the system, and determining customer satisfaction (Pressman, 2014).

Two major testing were done, unit testing and system testing. Unit testing was a test of single components or modules of the system. Functionalities like login, uploading images, disease prediction and recommendation generation were tested with different input scenarios to verify the accuracy and strength.

System testing was conducted to test the functionality and performance of the integrated system in general. This involved frontend, backend and AI interaction testing. To obtain an idea of the effectiveness and efficiency of the system in terms of response time to image inputs, performance testing was also performed.

The AI model's performance was evaluated based on prediction accuracy and confidence levels. The use of CNN-based models has proved to be very useful in the detection of plant diseases, with high accuracy recorded when the model is trained using quality data (Mohanty et al., 2016). However, it was observed that the model's accuracy depends on the diversity and quality of the training data.

Moreover, the usability testing was done with surveys to obtain user feedback. The difficulties encountered by farmers were brought to the fore by pre-survey results, whereas the post-survey results showed better usability and satisfaction with the system. Bugs and performance problems detected in the testing were fixed to improve the system reliability.

3.3.4. Review

The review stage was aimed at analyzing the overall performance, effectiveness and impact of the system once it was developed and tested. This step is critical in determining the strengths, weaknesses and future improvement opportunities (Sommerville, 2016).

Post-surveys were used to capture user feedback to evaluate the usability, accuracy and general satisfaction of the system. The findings indicated that the system was useful to the users in identifying the diseases in crops and in making decisions. The system greatly minimized time spent in identifying the disease and access to credible information.

The system was also critically analyzed to determine how it performs in a real world situation. Although the system showed good potential in identifying diseases, some limitations were discovered, including the reliance on internet connectivity and the lesser quality of detection on uncommon or novel disease. Such shortcomings indicate the necessity of further enhancement in the model training and system design.

According to the review, some recommendations to work out in the future were offered. They involve increasing the size of the dataset to enhance AI, supporting more types of crops, offline capabilities, and features of the user interface. Moreover, real-time data collection and analysis can also be enhanced by incorporating IoT-based monitoring systems.

On the whole, this review stage was a solid ground to smart agriculture solutions because it has guaranteed that the system achieved its goals and offered a lot of information to improve the system in the future.

3.4. Survey Results

3.4.1. Pre- Survey Results

An initial survey was carried out among 30 people to capture the existing practices, awareness and issues around crop disease detection. The survey consisted of farmers, students, and people representing other professions, most of the respondents were students (76.7%) and only a few of them were farmers. It means that, though direct participation of farmers is minimal, the survey still shows the general awareness and perceptions on agricultural practices.

The findings indicate that crop disease is a widespread problem, with 46.7 percent of the respondents having it occasionally, and 36.7 percent rarely. Frequencies were only reported by a

small percentage. This implies that although not permanent, crop diseases are a perennial issue in the agricultural sector.

Regarding the existing practice, majority of the respondents use traditional practices in identifying the disease. Approximately 56.7% of them consulted experts, 46.7% make use of the internet or such resources as YouTube, and 40% rely on the visual observation. A small percentage of respondents (6.7) said that they had no method at all. This means that detection of diseases remains to a large extent manual and relies on external support, which is not always available.

The survey also brings out awareness as far as the significance of early detection is concerned. Most people (53.3%), are of the opinion that crop loss can be minimized by early detection of the disease, 36.7% are not sure. This demonstrates a moderate amount of awareness yet also reflects the necessity of further education and trustworthy tools.

With technological familiarity, 56.7% of the respondents are familiar with mobile applications to some extent with only 16.7% indicating that they are very familiar. This implies that although solutions based on mobile are an option, the system should be easy and accessible enough to be adopted.

The main challenges in manual disease detection mentioned by the respondents were the inability to identify the symptoms (36.7) and the lack of knowledge (23.3), with a significant number of respondents (36.7) reporting all of them. This reinforces the need for an automated solution.

Moreover, new technology is of high interest to adopt. Only 53.3% of respondents indicated that they would definitively use an AI-based disease detection app, and 46.7% indicated that it would be used maybe, which suggests the potential to have a high number of users.. It was also found that accuracy was of great significance with 55.2 percent of the respondents indicating it as being extremely important.

Use of smartphones (63.3%), and offline mode (75.9) is a preference of most of the respondents which indicates important design requirements. Moreover, 63.3 per cent of the users like to use the application in English, whereas others like to use Nepali or local languages implying that the application should be multilingual.

Lastly, the issues related to the use of AI are the question of trust (63.3%), accuracy (56.7%), and cost (36.7%). Even though these are the concerns, the majority of the respondents think that AI can enhance agricultural output.

Altogether, the pre-survey outcomes evidently suggest the restriction of the existing disease detection solutions and reveal the necessity of an efficient, precise, and convenient AI-based solution.

3.4.2. Post- Survey Results

Following the implementation and testing phase of the AI-Powered Crop Disease Detection and Recommendation System, a post-survey was carried out with 30 respondents to assess the performance, usability, accuracy and effectiveness of the system in the real agricultural applications. The survey was designed to know user experience upon interacting with the system and to determine the effectiveness of the system in decision making in crop disease management.

The findings show a very good overall user experience. Forty-two point one percent of the users rated the system as excellent with another 42.1 percent rating it as good. It was rated as poor by none of the users and only 15.8% rated it as average. This is a clear indication that 84.2% of the users had a good experience with the system implying that the system is able to address the expectations of the users and offer a satisfactory solution to detecting crop diseases.

The system proved to be extremely user-friendly in terms of usability. Approximately 57.9% of users indicated that the system was easy to use and 26.3% indicated that the system was very easy to use. A smaller percentage of 15.8% rated it as moderate and none of the users had trouble using it. This proves that the majority of users (84.2 percent) found the system simple or very simple to use, which testifies to the fact that the interface is user-friendly and appropriate even to users with less technical background like farmers.

User feedback was also highly positive when it comes to the accuracy of detecting a disease. Approximately, 57.9% of users rated the system as accurate, 31.6% rated it as slightly accurate and 10.5% rated it as very accurate. Notably, there were no complaints of inaccurate results among the users. This suggests that the predictions of the system were deemed as reliable to a certain degree by all users, which demonstrates good performance of the CNN-based model in a real-life setting.

The usefulness of treatment recommendations was much valued as well. Most of the users (55.6) rated the recommendations as very useful with 38.9% as useful. Only 5.5 percent rated them as slightly useful and none of the users thought they were not useful. This implies that 94.5 percent of users believed the recommendations to be useful or very useful, indicating that the system offers valuable agricultural advice on other than disease detection.

Speed was also a favorable evaluation of system performance. The users rated the system to be fast by 52.6 percent, very fast by 15.8 percent, average by 26.3 percent and slow by 5.3 percent.

All in all, 68.4% of users were fast or very fast in performance, which means that the system is efficient to most users.

Regarding reliability and decision support, 52.6% of users had mostly confidence in the system, 10.5% fully trusted the system and 36.8 were not sure. This demonstrates that trust is usually a good thing but that additional enhancements in accuracy and awareness may lead to better user confidence.

The interface was also satisfactory to the user. Approximately 42.1 percent of the users were satisfied, 21.1 percent were very satisfied, and 36.8 percent was neutral with none of the users indicating dissatisfaction. This implies that 63.2 out of the users were satisfied or highly satisfied though there were some suggestions on improvements of the UI design.

Some of the user feedback that was common was recommendations to work on the interface design, minor technical problems, increase accuracy of rare diseases and have offline capabilities to be more accessible.

To sum up, the results of the post-survey of 30 users indicate clearly that the AI-Powered Crop Disease Detection System is efficient, reliable and easy to use. Most users have given good reviews on the accuracy, usability and usefulness, and this proves that the system is a workable tool to a real-life agricultural problem. By implementing some small changes in the interface design, performance optimization, and increasing the volume of data within the system, one can make it even more efficient and more trusted by users.

3.5. Requirement Analysis

Requirement analysis is a basic process in the system development lifecycle as it determines what the system should or should not do and how it should or should not perform. In the case of the AI-Powered Crop Disease Detection and Recommendation System, this step will ensure that the needs of users and technical requirements are clearly identified and translated into functionalities of the system. It serves as a blueprint to guide developers, designers and the stakeholders through the development process.

The requirements can be broadly divided into non-functional requirements and functional requirements, where the former defines the quality of the system and its performance standards, and the former defines the basic features of the system.

3.5.1. Functional Requirement

Functional requirements outline the necessary services and operations that the system should offer to the users. These requirements are directly concerned with the interaction of users with the system and the response the system gives to user inputs.

- **User Authentication**

The system should have a well-authenticated system that enables users to create accounts, log-in, and manage them. This aspect guarantees that system functionalities and personal data are accessed by authorized users only. It also enables personalization, such as storing user history and preferences. Secure authentication methods such as password encryption and session management are important in ensuring the integrity of the system and user trust.

- **Image Upload and Disease Detection**

One of the most important functions of the system is that users can upload images of crop leaves via an easy-to-use interface. After an image has been uploaded, it is processed by a system based on the Convolutional Neural Network (CNN) model that has been trained on large datasets of plant diseases. The model can extract characteristics like color, texture, pattern of the picture in order to determine the disease. The automated detection minimizes the use of manual inspection and gives more accurate and quicker results.

- **Display of Results / Recommendation Delivery.**

Once the image has been processed, the system should be able to show the results in a straightforward way. These are the identified disease name, the confidence level to show the accuracy of prediction and the severity of the disease. Moreover, the system must also give practical recommendations like treatment procedures, prevention, and recommended farming practices. This is to guarantee that the system does not just identify the problems but also assists users in solving them.

Data Management and History Tracking: monitors, tracks, and documents records (Kohki, 2001). The system must also have a formalized database to contain the user information, uploaded images, and prediction outcomes. This allows users to see their previous history and see how disease trends have changed over time. Past data can assist in finding common problems and assessing the effectiveness of the treatments, therefore helping to make more effective long-term decisions.

- **API Integration**

RESTful APIs need to be deployed to guarantee a smooth flow of information between frontend interface and the processing system in the back-end. These APIs manage the exchange of data, such as sending pictures to be analyzed, obtaining the results of the predictions, and storing user

information. This integration enables the system to run smoothly and offer a user friendly experience.

3.5.2. Non- Functional Requirement

The non-functional requirements are used to specify the effectiveness of the system in carrying out its functions. These requirements play a major role in quality of the systems, satisfaction of the users and the long term sustainability.

- **Performance**

The system should be able to give quick and correct results with the ideal results being the ability to process images and give predictions in a matter of few seconds. It must also be in a position to support multiple users at the same time without delays. Scalable cloud infrastructure has been used to ensure that there is consistency in performance even when it is in high demand.

- **Usability**

The system should be simplified and made easy to understand since most of the people who will be using the system are farmers who might lack the technical knowledge. The interface must be user-friendly and not a lot of actions are needed to post images and see the results. It should have clear instructions, readable text, and visual aids to facilitate user understanding and accessibility.

- **Reliability and Availability**

The system must be stable and with a low downtime. It should be accessible when required by the users particularly in periods of critical farming. The system must also be able to gracefully deal with errors and to recover fast in case of failure to maintain seamless service.

- **Security**

It is critical to ensure the safety of user data. The system should provide high security measures like data encryption, secure systems of login and safeguarding against unauthorized access. This will guarantee secrecy and user trust in the system.

- **Maintainability and Extensibility**

The system must be in a modular and flexible way so that it can be easily updated and enhanced in the future. It involves the introduction of new disease detection models, serving more types of crops, and incorporating new features. A sustainable system saves on the overall expenses in the long run and is flexible to meet the changing agricultural demands.

Requirements Analysis Conclusion.

The requirement analysis in general will give a detailed picture of the functional capabilities of the system as well as the quality expectations of the system. It makes sure that the AI-Powered Crop Disease Detection and Recommendation System does not only fulfill its purpose but also provides high performance, security, and usability. This systematic definition is a powerful background of the design, development, testing and deployment processes that will ensure the successful implementation of the system.

3.6. Design

The AI-Powered Crop Disease Detection and Recommendation System design phase converts the system requirements into a hierarchical technical design. It specifies the interaction between the system components, data organization and implementation of functionalities. The key elements of the design are the UML diagrams, entity relationship diagram (ERD), and system architecture. All these are to provide clarity, modularity, scalability, and maintainability of the system.

3.6.1. System Architecture

The system architecture is an essential roadmap, which determines the structure of the system and how various parts of the system will relate to each other to realize the desired functionality. In the present project, we have implemented the layered architecture, which splits the system into distinct interdependent layers. This will guarantee modularity, scalability, maintainability, and clarity as well as provide an avenue to enhance or upgrade in future without interfering with any existing functionality (Shocodevin, 2025).

Functionality and Layer Breakdown.

- Frontend Layer
The interface layer is the interface point between the system and the users who are farmers, the agricultural experts and the administrators. This interface is meant to be simple and easy to use, enabling farmers to post photos of crops, get predictions of diseases, and have treatment plans. AI predictions can be validated by experts and guided, and knowledge bases can be updated, and admins can be able to monitor the overall system activity. This way, we can guarantee that all modifications to how the system is being displayed, including redesigning forms or dashboards, does not affect the functionality of the backend code or AI processing (Prepbytes, 2023).
- Backend Layer

The backend layer is a system layer that handles the business logic of the system. It receives the user requests in the front end and interprets them based on system rules and sends messages to the AI service and the database. An example would be when a farmer sends a crop picture, the backend will validate this input, send it to the AI layer to analyze it and the prediction results will be stored back to the rear of the frontend and displayed. The location of business logic in the backend allows the processing of all data to be consistent, secure and maintainable (Tpoint Tech, 2025).

- **AI Service Layer**

The layer of AI service will involve making intelligent decisions in the system. It uses machine learning algorithms, including Convolutional Neural Networks (CNNs), to identify crop diseases and provide the correct treatment options. When an image has been passed down the backend to this layer, the AI model works on it, identifies possible diseases, and returns actionable information. It is possible to enhance the models, add more training data, or add predictive analytics, and leave other system layers unchanged by isolating the AI services into a separate layer (Shocodevin, 2025).

- **Layer**

The database layer is a layer that is focused on the storage and management of structured data. These are user profiles, crop data, disease history, prediction history, and analytics outcomes. This system will allow the storage and efficient querying of diverse information, as it will manage data in a distinct layer, and allow scaling that can be used as the number of farmers and crop data grows. This division also permits the AI as well as the backend layers to run without having to engage directly with the raw data storage, enhancing performance and reliability (Wikipedia, 2025).

The benefits of the Layered Architecture.

- i. The layered approach has a number of advantages:

- **Maintenance:** Developers are able to work on a single layer without impacting the other thus minimizing errors and making the process of testing less complicated.
- **Scalability:** It is possible to scale the use of a specific layer to increase the number of users, crops, or AI models as required. As an illustration, when an increasing number of farmers are on the platform at the same time, the back-end and database may be scaled without altering the frontend or AI models.
- **Upgrades in the Future:** It can be upgraded to accommodate new AI algorithms or new features or redesign user interfaces without involving the system itself, which makes the system flexible and adaptable to future demands.

Attainable Workflow.

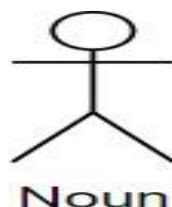
As an example, in case a farmer uploads a leaf image, the request is received by the frontend layer which gives a confirmation interface first. The picture is then forwarded to the backend where it is validated and preprocessed. Then, the AI service layer interprets the image to identify diseases and provide recommendations. Lastly, the predicted value and the history of the interactions are stored in the database layer and the results are returned back on the backend to the frontend so that the farmer may see. This is an illuminating flow of the importance of each layer in the process of the smooth, precise, and efficient functioning.

3.6.2. Use Case Diagram

A Use Case Diagram is a visual representation of the functional requirements of a system, highlighting the interactions between **actors** (users or external systems) and the system itself. It captures what tasks the system can perform, who can perform them, and how the system responds to different user actions (Sommerville, 2016). In the AI-Powered Crop Disease Detection and Recommendation System, the use case diagram demonstrates the dynamic interactions between farmers, administrators, agricultural experts, and the AI system. Farmers can submit images of crops to detect diseases, receive automated treatment recommendations, and track the health of their crops over time. Experts can validate disease diagnoses and update treatment suggestions. Administrators manage user accounts, monitor system performance, and ensure data integrity. The AI system acts as a central processing entity, analyzing submitted crop images, detecting diseases using machine learning models, and generating relevant recommendations for treatment. This use case representation helps stakeholders understand the system's functional capabilities, clarifies user roles, and serves as a foundation for system design and development (Jacobson et al., 1992).

3.6.2.1. Notation used in use case

i. Actor



Actors give the various kinds of users/external systems expected to communicate with the system. They may be individuals, other applications or computer hardware that interact with the system to get things done. The system presents certain roles to different

actors and each communicates with the other to reach a goal. Actors in a use case diagram are usually depicted as a stick figure outside the system boundary. As an example, a customer, administrators, and sellers are the actors likely to appear in an e-commerce system. The naming convention is noun.

ii. Use Case

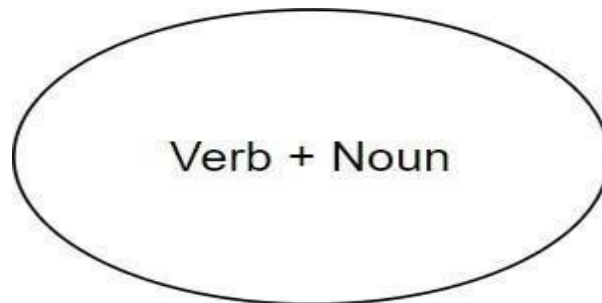


Figure 4: Notation for Use Case

Describing how the system responds to actor requests, use cases detailed the actual actions/tasks performed by the system in response to such requests. Every use case symbolizes a functional requirement, or a specific service that the system provides to the persons. Within the system bound, use cases are shown as horizontal ovals (or ellipses) with the name of the functionality written in them. As an example, use cases could consist of the following in the same context of the e-commerce example: such as Register Account, Place Order, Manage Products, and Generate Sales Report. These assist in explaining what the system should perform on the part of the users.

iii. System boundary

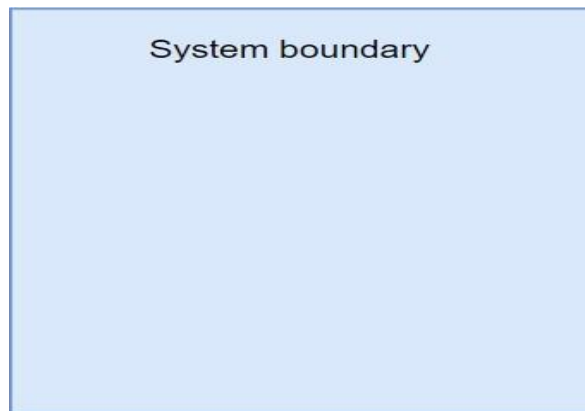


Figure 5: Notation for System boundary

The system boundary controls what is within a system and excludes actors outside a system. It is signified by a great rectangle (box) that contains all the use cases and depicting what belongs to the system or what is outside of it. This is because usually the name of the system is written in the top part of the rectangle. The presented visual representation assists the stakeholders to make a clear distinction between internally defined behaviors and externally defined behaviors and thus have a well-scoped system design.

iv. Association

Figure 6: Notation for Association

An association is shown by a solid straight line between an actor and a use case. It means that the actor is involved in the use case, either as a producer of the call that leads to use case or as a consumer of some results of the use case. It is in most cases that they do not have an arrowhead as their lines unless the direction is required to be indicated. The use case diagram depends on associations to resemble its foundation based on the direct interaction of the system and the user (or external element).

v. Include

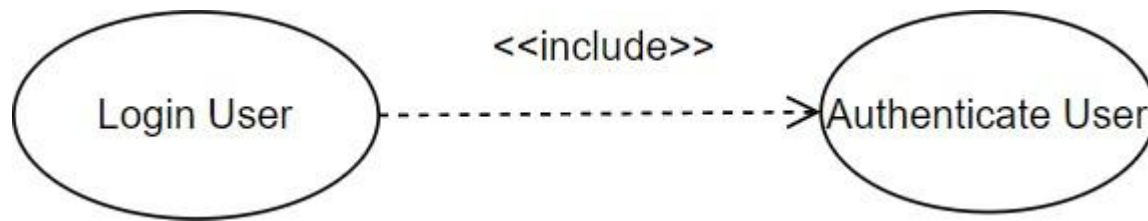


Figure 7: Notation for Include

The include relationship is represented by a dashed arrow labelled «include» and whose direction is drawn as showing the base use case to the included use case. It refers to the fact that the behavior of the included use case will be always executed when executing the base use case. This prevents duplication as it is possible to share functions. In diagrams, the arrow is extended towards the inclusion use case and indicates that such an inclusion is mandatory.

vi. Extends

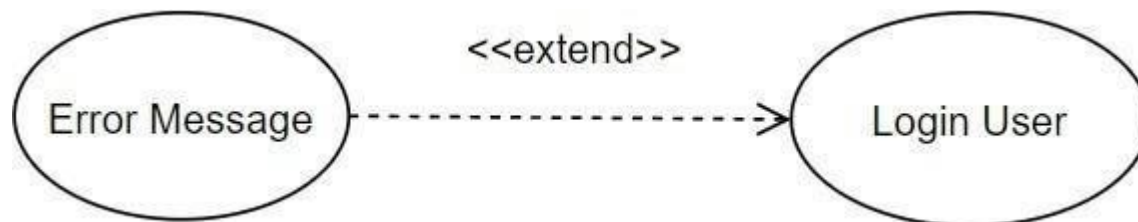


Figure 8: Notation for Extends

The extend association is also illustrated using a dashed arrow but the direction is in opposite as labelling whereby the label is extended and the extended use case points at the base use case. It describes optional or contingent behavior, which can be activated, only under some conditions. It means that the base use case can remain short and extensions can be addressed individually. The direction of the arrow, against which optional functionality, directs to the major covering use case.

vii. Generalization



Figure 9: Notation for Generalization

Generalization is shown as a solid line having a hollow triangle arrow at the end with the more specific actor or use case on the tip. To actors, this implies that all the roles and responsibilities of the parent actor are transferred to the child actor. To be used, it indicates that the specialized use case is the special form of the generalized one. The visual provides reuse of actors/behaviors and hierarchy of actors/behaviors

3.6.2.2. Actor and Use Case

1. Actors:

- Farmer
- Admin
- Expert
- AI System

2. Use Cases:

i. Farmer:

- Register
- Login
- Upload Crop Image
- View Disease Detection Result
- View Confidence Score
- View Treatment Recommendations
- View Prediction History
- Receive Disease Notifications
- Participate in Discussion Forum

ii. Admin

- Manage Users
- Manage Crop Disease Dataset
- Retrain AI Model
- View Analytics Reports

iii. Expert

- View Farmer Queries
- Reply to Farmer Queries

iv. AI System

- Preprocess Image
- Detect Crop Disease
- Store Prediction History
- Generate Analytics
- Send Notifications

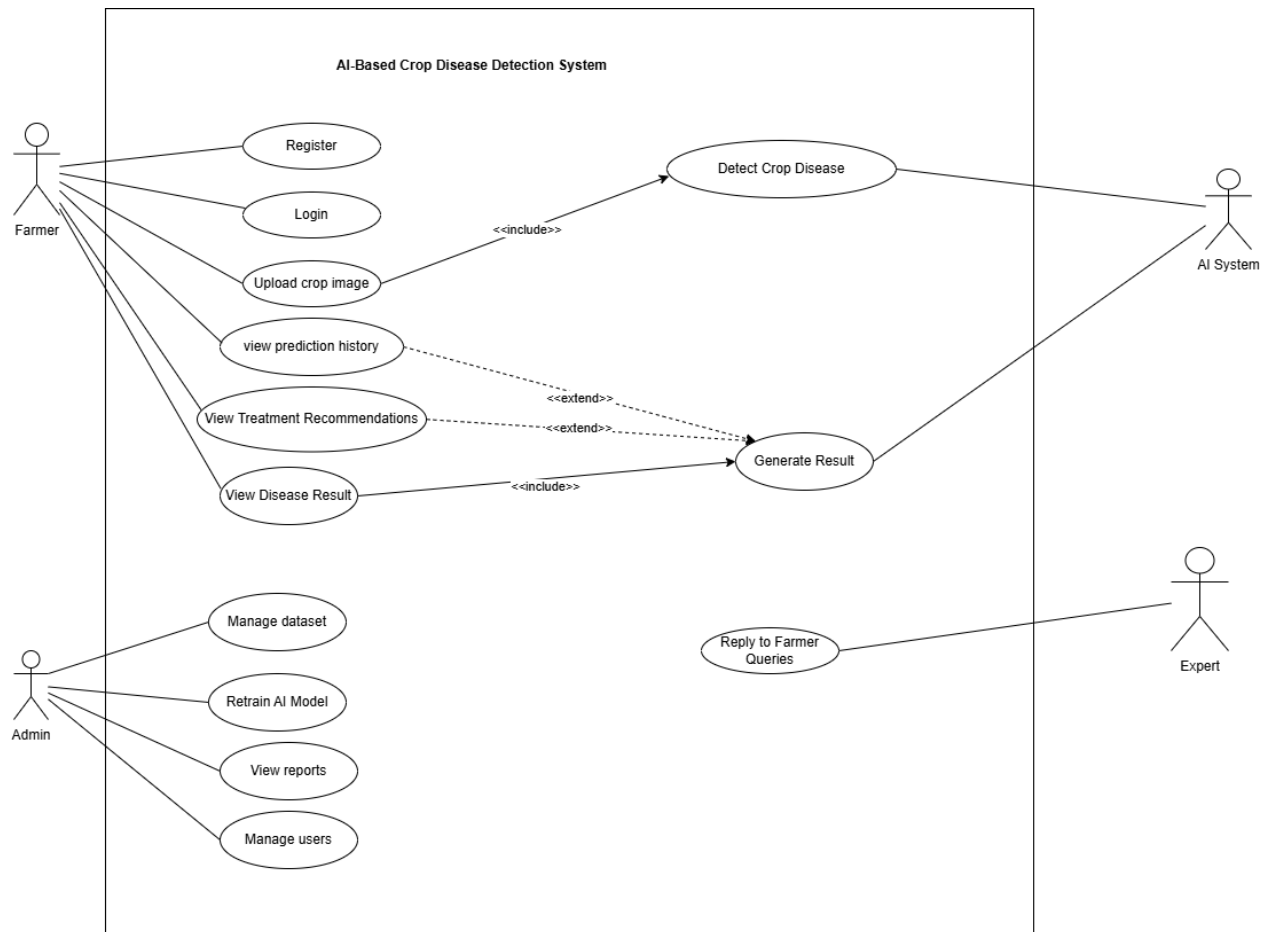


Figure 7: Usecase diagram

Aligned User Stories:

US001, US002, US003, US006, US007, US008, US011, US012, US015, US016, US017, US022, US023, US024

This use case diagram illustrates the interaction between farmers, administrators, experts, and the AI system. It highlights the functional requirements of the system, including disease detection, treatment recommendation, dataset management, and expert interaction.

3.6.3. Activity Diagram

An Activity Diagram is a behavioral diagram in UML that represents the workflow of activities within a system. It shows the step-by-step progression of control from the initiation of a process to its completion, illustrating the sequence of actions, decision points, parallel activities, and synchronization between different tasks (Booch et al., 2005). Activity diagrams are particularly

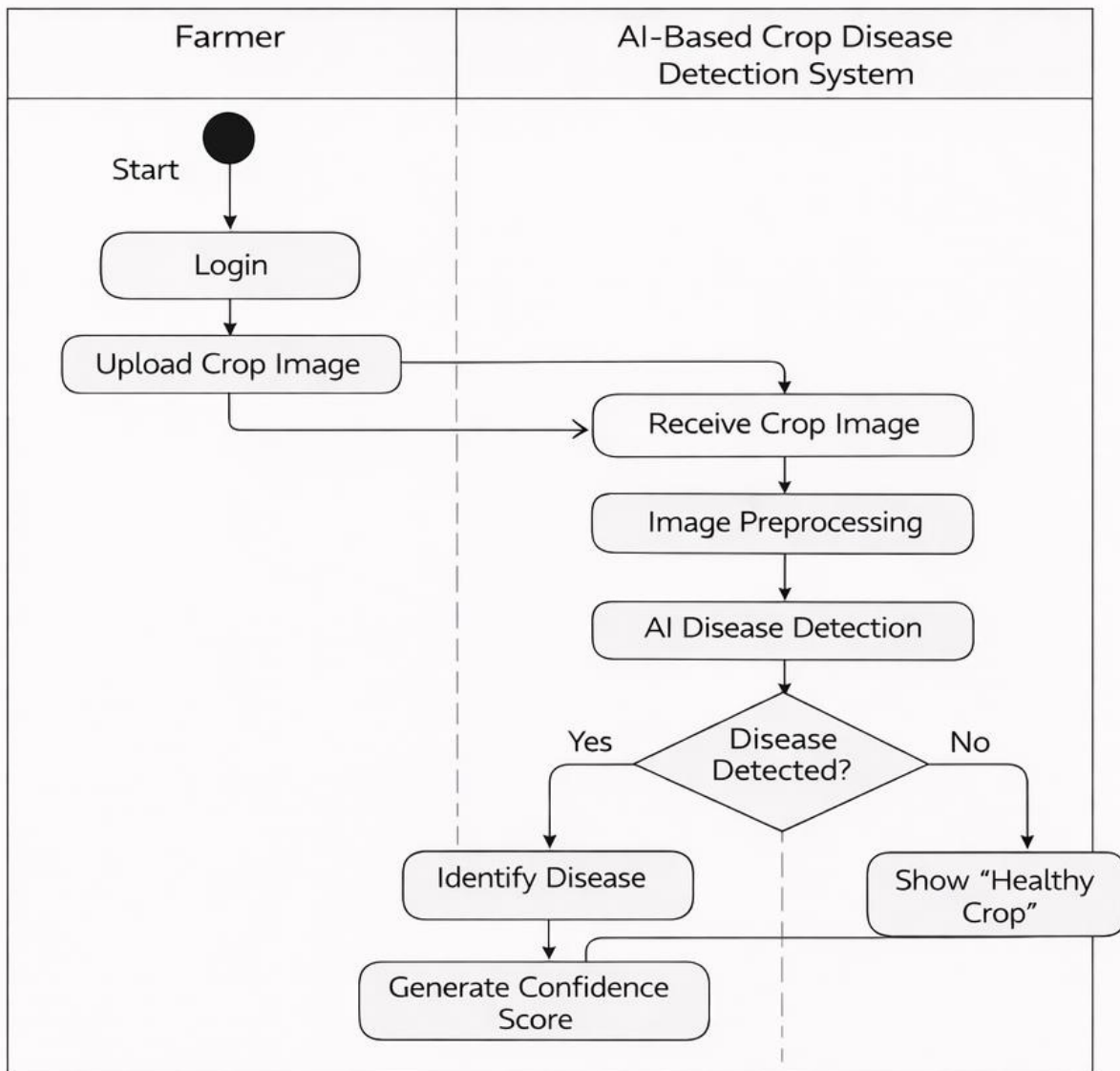
effective for modeling business processes, system workflows, and use case realizations, as they visually clarify how various activities are executed in response to specific events. They provide stakeholders with a clear understanding of the dynamic behavior of the system by illustrating how actions are coordinated and how control transitions between them.

In the AI-Based Crop Disease Detection and Recommendation System, an activity diagram models the workflow beginning with image upload by the farmer, followed **by** image preprocessing, disease detection using the AI model, result generation, and finally the display of treatment recommendations. By representing these steps, the activity diagram helps ensure a clear understanding of system operations, supports effective system design, and enables validation of the processes, thereby enhancing both usability and reliability of the system (Ambler, 2004).

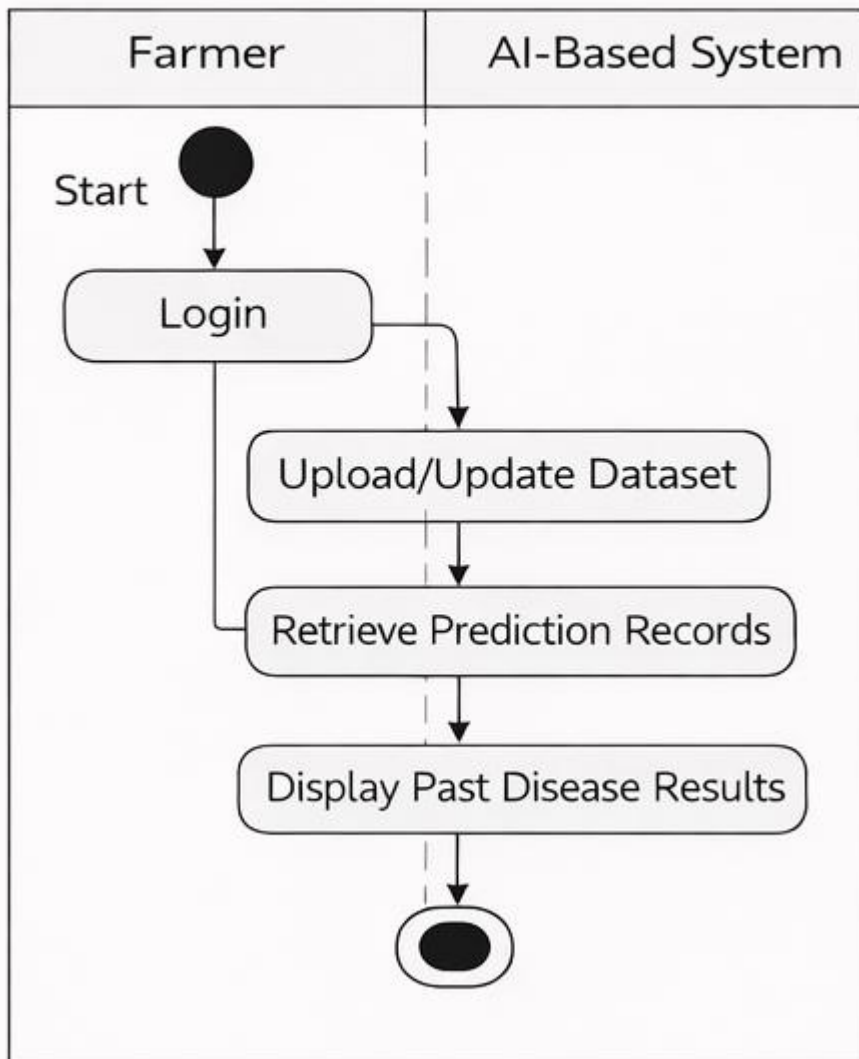
Table 4: The Notation of Activity diagram

Sr. No	Name	Symbol Description
1.	Start Node	 (Filled black circle)
2.	Action State	 (Rounded rectangle)
3.	Control Flow	 (Arrow line)
4.	Decision Node	 (Diamond shape)
5.	Fork	 (One line splitting into multiple arrows)
6.	Join	 (Multiple lines joining into one line)
7.	End State	 (Black circle inside a white circle)

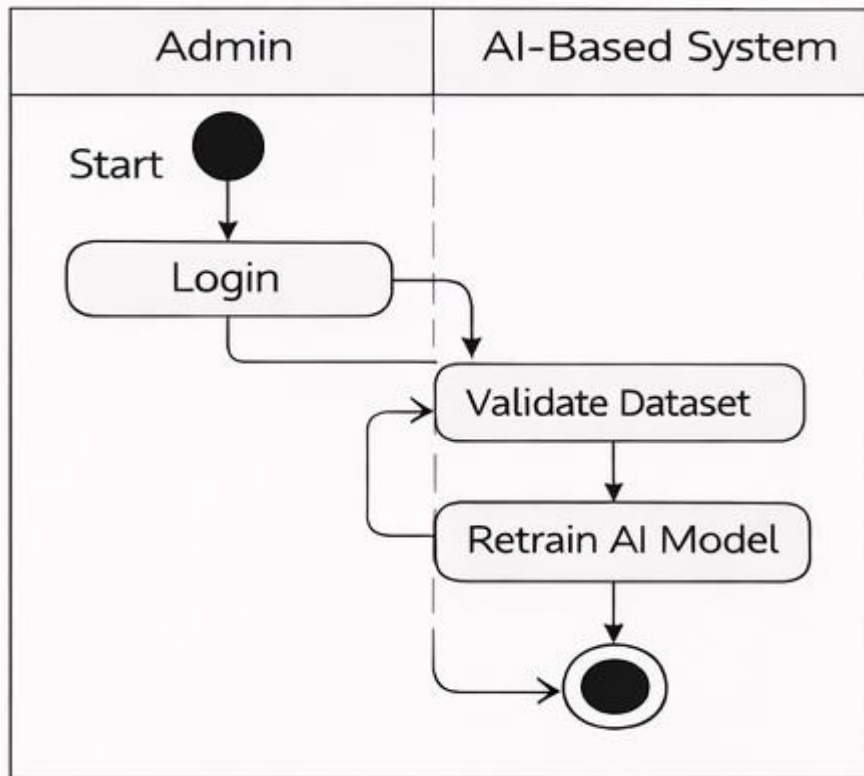
Crop Disease Detection Process



Prediction History Management

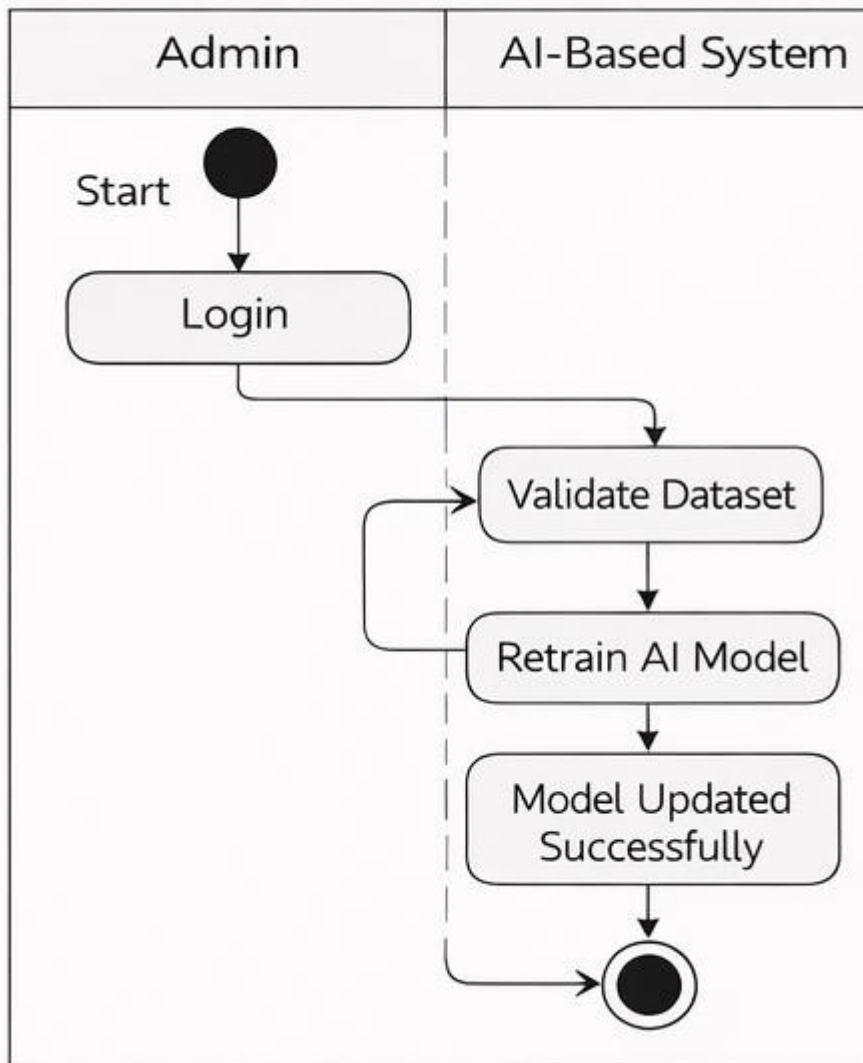


Model Retraining

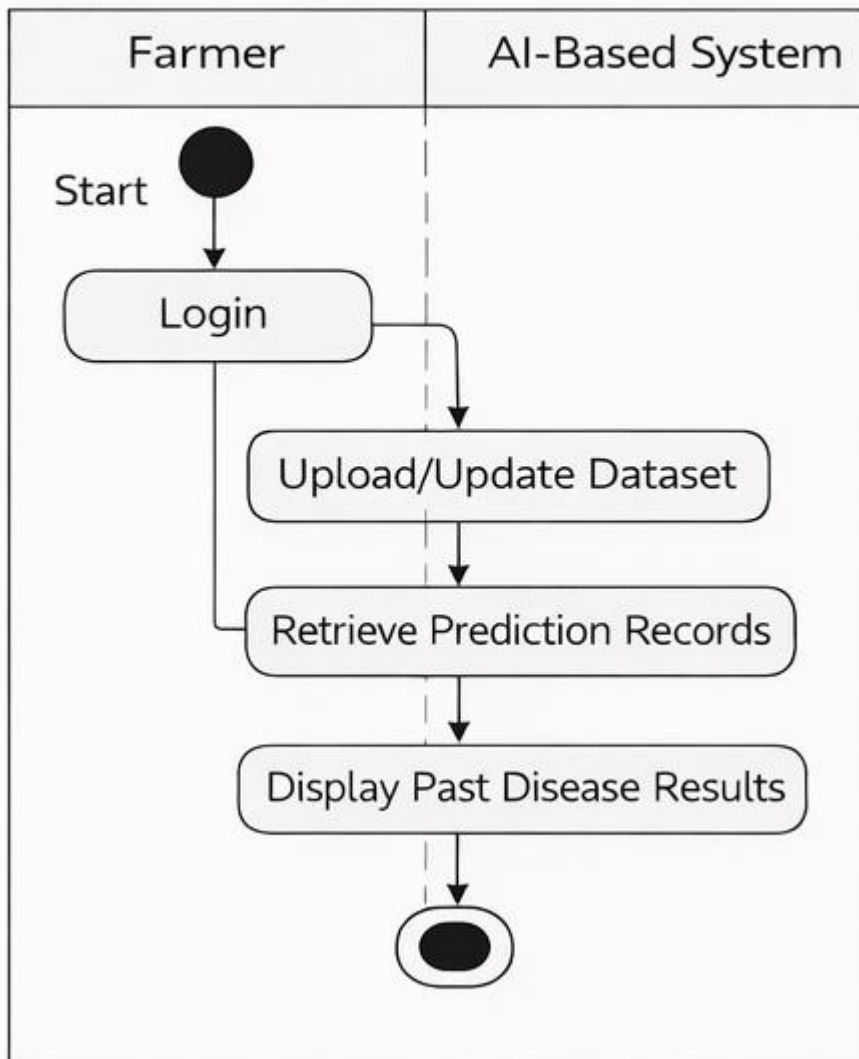


Forum Support

Model Retraining



Prediction History Management

**Aligned User Stories:**

US003, US004, US005, US006, US007, US008, US011

3.6.4. Sequence Diagram

A Sequence Diagram is a type of UML interaction diagram that illustrates how objects or system components interact with each other over time. It focuses on the order of messages exchanged between different entities to accomplish a specific functionality or use case. The diagram presents participants as lifelines arranged horizontally, while messages are shown as arrows that indicate the direction and sequence of communication from top to bottom, representing the flow of time (Booch et al., 2005).

Sequence diagrams are particularly useful for modeling dynamic behavior, validating system logic, and understanding how responsibilities are distributed among system components. They help developers and stakeholders visualize system processes, identify dependencies, and ensure that interactions occur in the correct order. By clearly depicting method calls, responses, and control flow, sequence diagrams support effective system design and improve communication among development teams (Sommerville, 2016).

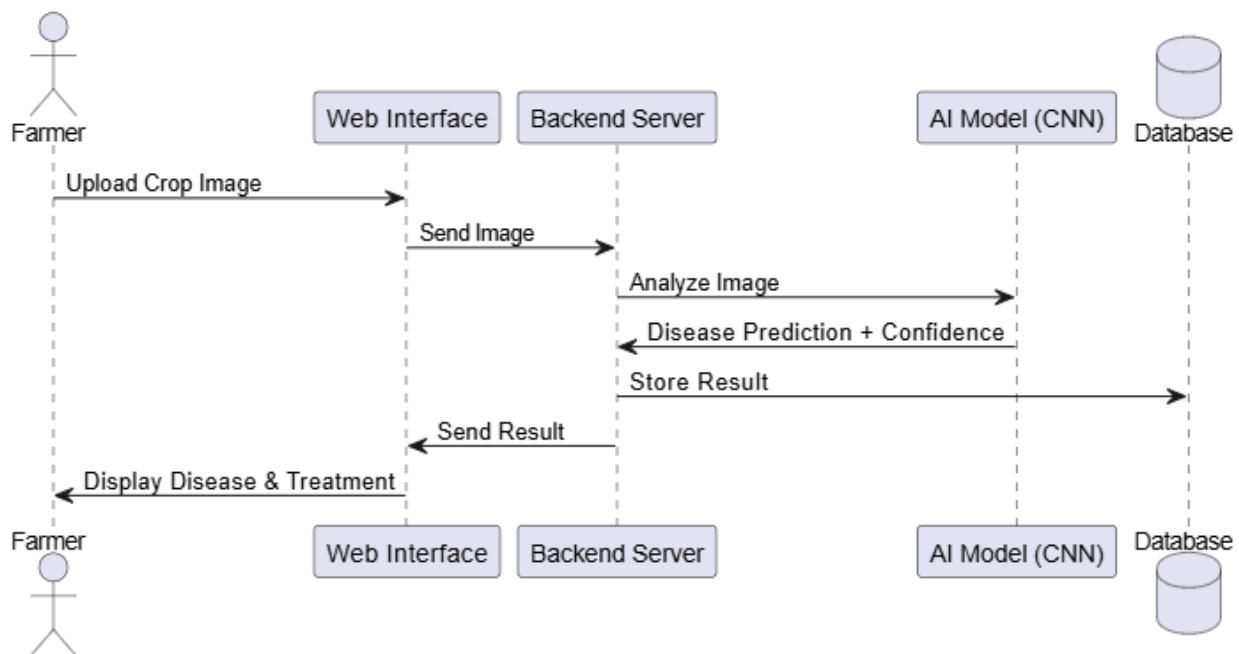


Figure 8: Sequence Diagram

3.6.5. Entity-Relationship Diagram (ERD)

The ERD for the AI-Powered Crop Disease Detection and Recommendation System presents the structural layout of the database layer supporting all the AI-driven prediction operations, user roles, farming knowledge-base storage, community interaction, and system analytics. Each entity defines a data object in the system along with its properties, and relationship mapping ensures seamless

workflow between prediction, treatment suggestion, calendar management, and expert advisory support.

The architecture is scalable and supports various user categories, model upgrades, dataset retraining, automated notifications, and role-based data access.

3.6.5.1. Entities and Attributes

In a database system, entities represent the primary objects or components about which data is stored. These can be people, objects, events, places, or concepts that exist within the system and are of some significance to the application. Each entity will be considered as an independent data unit, and in a relational database, it is normally mapped to a table. For instance, in an AI-powered crop disease detection system, major data elements such as User, Crop, Disease, Farmer, and AI_Model are the entities that the system interacts with and maintains information regarding.

Attributes, on the other hand, describe characteristics or properties of each entity. They define what type of information is stored within an entity, and in a database, they would be represented as fields or columns within a table. Each entity contains multiple attributes in order to uniquely define its behavior and structure. Such an example could be the User entity, which might have full_name, email, contact_number, and role as its attributes. These define the particular identity and details of any user. Put simply, entities describe what the system is storing data about, while attributes describe what detail of that entity is being stored.

1. User

user_id (PK)
full_name
email
password_hash
contact_number
role
date_created

2. Admin

admin_id (PK)
user_id (FK)
permissions
last_login
created_at

3. Expert

expert_id (PK)
user_id (FK)

specialization
certification_id
verified_status
rating

4. Farmer

farmer_id (PK)
user_id (FK)
location
land_size
preferred_crop

5. Crop

crop_id (PK)
crop_name
type
growth_duration
suitable_climate
soil_type

6. Disease

disease_id (PK)
disease_name
crop_id (FK)
severity_level
description
common_symptoms

7. AI_Model

model_id (PK)
model_version
training_dataset
accuracy_score
deployment_status
updated_at

8. Prediction

prediction_id (PK)
user_id (FK)
disease_id (FK)

confidence_score
image_path
prediction_timestamp

9. Treatment_Recommendation

recommendation_id (PK)
disease_id (FK)
treatment_type
product_name
dosage_instruction
eco_friendly_option

10. Fertilizer

fertilizer_id (PK)
fertilizer_name
crop_id (FK)
soil_requirement
application_method
recommended_season

11. Pesticide

pesticide_id (PK)
product_name
chemical_composition
application_frequency
safety_precautions
disease_id (FK)

12. History_Record

record_id (PK)
user_id (FK)
prediction_id (FK)
status
notes
record_created

13. Crop_Calendar

calendar_id (PK)
crop_id (FK)
planting_month

harvesting_month
season_alerts

14. Notification

notification_id (PK)
user_id (FK)
message
notification_type
status
created_at

15. Forum_Post

post_id (PK)
user_id (FK)
title
content
image_path
created_date

16. Forum_Reply

reply_id (PK)
post_id (FK)
user_id (FK)
reply_content
created_at
upvotes

17. Dataset

dataset_id (PK)
source_name
total_images
labels_count
last_updated

18. Analytics_Report

report_id (PK)
admin_id (FK)
report_type
generated_date
affected_region

key_metrics

19. Region

region_id (PK)

region_name

climate

humidity_level

soil_ph

risk_index

20. Subscription

subscription_id (PK)

farmer_id (FK)

plan_type

payment_status

valid_until

3.6.5.2. Relationship Mapping**i. User Role Relationship**

A User account can exist as a Farmer, an Expert, or an Admin

Type: Inheritance

Cardinality: 1 User = 1 specific role

ii. Crop & Disease

One Crop may have multiple associated Diseases

Type: One-to-Many

Cardinality: Crop (1) → Disease (Many)

iii. Disease & Recommendations

One Disease can have several Treatment Recommendations

Type: One-to-Many

Cardinality: Disease (1) → Recommendation (Many)

iv. Disease & Pesticides

A Disease may require more than one Pesticide product

Type: One-to-Many

Cardinality: Disease (1) → Pesticide (Many)

v. User/Farmer & Prediction

A single user (or specifically a farmer) can upload multiple Predictions

Type: One-to-Many

Cardinality: User (1) → Prediction (Many)

vi. Prediction & Disease

Each Prediction corresponds to one Disease result

Type: Many-to-One

Cardinality: Prediction (Many) → Disease (1)

vii. Prediction & History Record

Every Prediction has one and only one History Record

Type: One-to-One

Cardinality: Prediction (1) → History Record (1)

viii. Crop & Crop Calendar

Each Crop maintains one Crop Calendar schedule

Type: One-to-One

Cardinality: Crop (1) → Crop Calendar (1)

ix. User & Notifications

One User may receive many Notifications over time

Type: One-to-Many

Cardinality: User (1) → Notification (Many)

x. User & Forum Posts

A User is allowed to write multiple Forum Posts

Type: One-to-Many

Cardinality: User (1) → Post (Many)

xi. Post & Replies

A single Post can have several Replies from users

Type: One-to-Many

Cardinality: Post (1) → Reply (Many)

xii. Dataset & AI Models

One Dataset can be used to train multiple AI_Model versions

Type: One-to-Many

Cardinality: Dataset (1) → Model (Many)

xiii. Admin & Reports

Each Admin can generate multiple Analytics Reports

Type: One-to-Many

Cardinality: Admin (1) → Analytics Report (Many)

xiv. Region & Farmers / Predictions

One Region can be associated with several Farmers and multiple Predictions

Type: One-to-Many

Cardinality: Region (1) → Farmer/Prediction (Many)

xv. Farmer & Subscription

A Farmer may have one active Subscription plan

Type: One-to-One

Cardinality: Farmer (1) → Subscription (1)

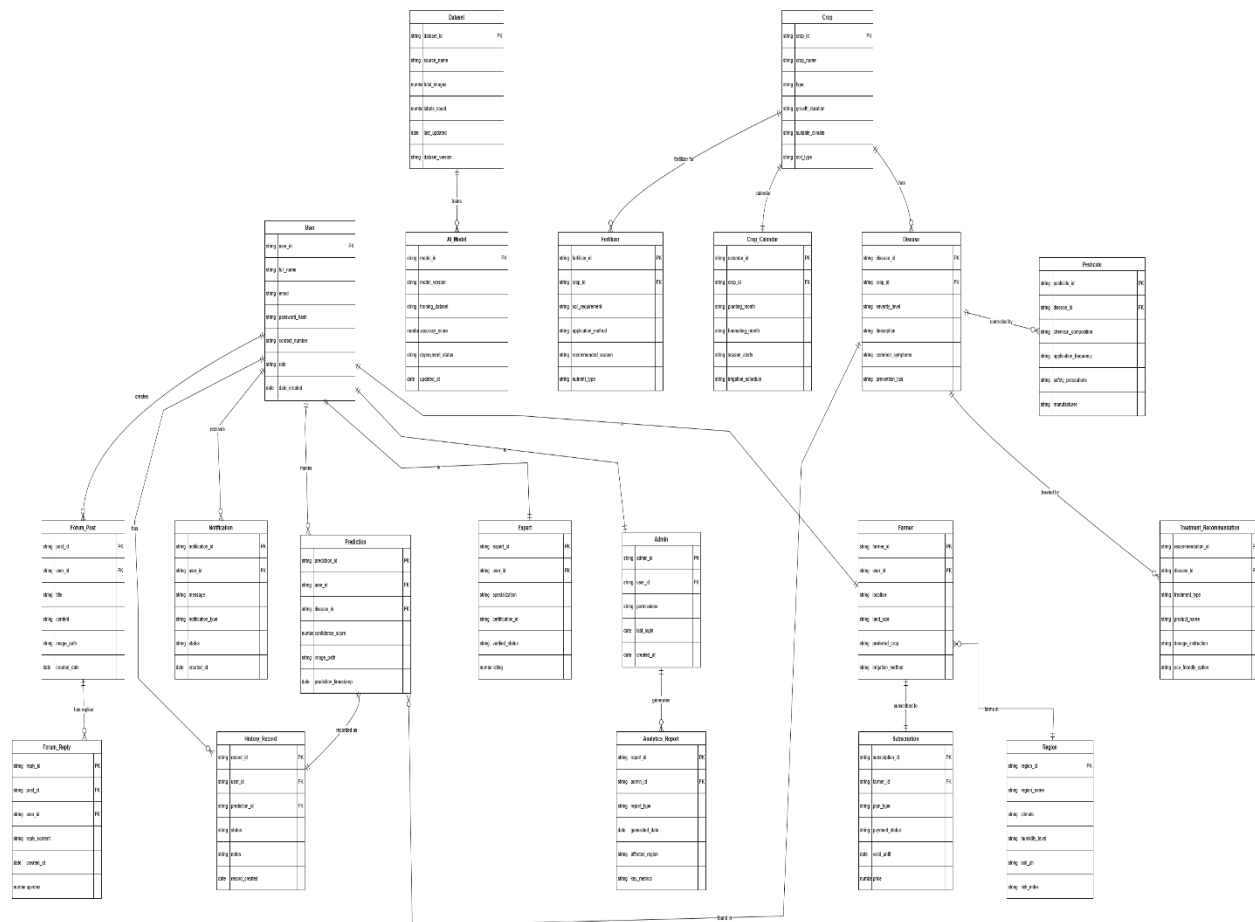


Figure 9: ERD

3.6.6. System Design

The system design phase is a foundational stage in software development where the project's requirements are transformed into a structured framework that directs how the system will be built. This phase goes beyond simply listing features; it defines *how* the system's components will work together, how data will flow among them, and how different users will interact with the system overall. In effect, system design creates a blueprint that guides the development team from concept to implementation, reducing ambiguity and helping prevent costly mistakes later in the project lifecycle (Prepbytes, 2023; Shocodevin, 2025).

A well-designed system does not emerge by accident it is the result of careful planning that balances clarity, modularity, and scalability, which are crucial quality attributes:

- **Clarity:** At its core, clarity in system design ensures that every component and interaction within the system is well defined. A clear design helps developers and stakeholders understand exactly how different parts of the system relate to one another and what role each component plays. This shared understanding eliminates confusion during

implementation and makes it easier to communicate design decisions among team members. Architectural design serves as this clear roadmap by specifying the overall structure, components, and communication paths in a system (Tpoint Tech, 2025).

- **Modularity:** Modularity means dividing the system into smaller, independent units or modules such as frontend, backend, AI services, and databases — where each module handles specific functionality. This separation makes complex systems manageable by isolating responsibilities, lowering interdependencies, and allowing developers to work on individual modules without risking unintended impacts on the entire system. Modularity also improves maintainability because changes in one module do not necessitate widespread changes elsewhere (Wikipedia, 2025).
- **Scalability:** Scalability refers to the system's capacity to handle growth in terms of users, data, and operations without significant performance degradation. Designing for scalability means anticipating future increases in workload and choosing architectural patterns that support expansion without requiring a complete redesign. This is crucial for systems expected to serve many users or process large volumes of data, such as agricultural platforms used by thousands of farmers. Scalability ensures the system can adapt and evolve over time while maintaining performance and reliability (Shocodevin, 2025; Wikipedia, 2025).

By focusing on these principles during the system design phase, the project gains a solid technical foundation that aligns with both immediate requirements and future goals. A detailed design helps the development team plan effectively, improves overall communication, and ensures the system can be extended, maintained, and scaled with minimal friction as new features are added or usage increases (Prepbytes, 2023; Tpoint Tech, 2025)

3.7. Implementation

Implementation phase is the phase where the system design is translated into a complete functional software application. In this project, AI-Powered Crop Disease Detection and Recommendation System is deployed on a multi-layer architecture that includes the frontend, backend, and machine learning. The layers are built separately and combined to provide a smoother and efficient communication and performance. The system is based on a modular design, enhancing its scalability, maintainability, and its future enhancement.

3.7.1. Frontend Implementation

The AI-Powered Crop Disease Detection and Recommendation System frontend is implemented in React.js, a popular JavaScript library of creating interactive and responsive user interfaces. The primary objective of the frontend is to ensure a simple, efficient, and user-friendly interface that will enable users (particularly farmers) to interact with the system without technical skills.

With this type of system, the frontend is geared towards usability, accessibility and responsiveness. The interface is kept simple and user-friendly as the target users can be people with rural backgrounds and limited digital experience. React.JS enables the creation of reusable building blocks of the UI, which serves to ensure consistency throughout the application and less complexity in the development.

There are a number of critical modules in the frontend. First, the user authentication interface allows users to create and log in safely. This will guarantee every user a unique experience and their information is secured. Second, the image upload feature enables users to post crop leaf pictures with ease by use of the drag and drop or file selection feature. This component assists in supporting the basic image formats such as JPG and PNG.

The dashboard interface serves as the hub of the application. It enables users to manoeuvre around various functions which include posting new images, perusing previous predictions and receiving treatment advice. React elements (cards, forms, and navigation bars) are used to design the dashboard to make sure that it is organized.

The other important feature is the results display module, in which disease prediction results are indicated. This contains disease name, level of confidence and degree of severity. The results are presented in a graphical way to make sure that the users can easily comprehend the output without being technical.

Also, the frontend incorporates the recommendation section that refers to the treatment suggestions and preventive measures in a well-structured and readable format. This assists farmers to make quick decisions.

State management system of React is applied to manage dynamic updates of data and provide a seamless interaction without page reloads. The frontend interacts with the back-end through RESTful APIs, enabling the flow of data, i.e. transmission of images and getting predictions, to be seamless.

In general, the frontend is very important in user interaction, usability, and accessibility of the system, which makes the system useful in the real-world scenario in agriculture.

3.7.2. Backend Implementation.

The back-end of the system is coded with Node.js and Express.js framework. It serves as the main processing unit of the whole system, dealing with communication between the frontend, database, and AI model. The backend will process the business logic, user requests, and provide a secure and efficient data flow.

API request handling is one of the most important tasks of the backend. Each time a user post an image or carries out any activity on the frontend, the request is dispatched to the backend via RESTful APIs. These requests are processed in the backend and the data are checked and the right action taken.

User authentication and authorization is also taken care of by the backend. A secure system of logins and registration is introduced so that only authorized users can access the system. Encryption of passwords is done prior to storage to strengthen security and safeguard privacy of users.

Image handling and forwarding is another important role. As the user uploads a crop image, the backend temporarily stores the image or processes the image and sends it to the AI model to identify the disease. After the AI model provides the response, the backend acts on the response and sends it to the frontend.

The backend also plays a major role in database management. MongoDB is the database system that will store user information, pictures uploaded, history of predictions and system logs. MongoDB is selected as it offers the flexibility of working with unstructured data, which works well with image-based applications.

Also, the backend provides integration and flow of communication between components of the system. It serves as an interface between the frontend and AI model and makes sure that the data are transferred safely and effectively.

To guarantee stability of the systems, error handling mechanisms are also employed. In case of any failure in prediction or data processing, the backend will gracefully handle the failure and give the user the right feedback.

In general, the backend is the heart of the system, which guarantees the effective functioning of the system, safe data processing, and effective interaction of all modules.

3.7.3. AI Model

The heart of it is the AI model, which is implemented with deep learning methods based on Python. Convolutional Neural Networks (CNN) are used to construct the model and are very effective in image classification. CNNs find wide applications in medical imaging and agricultural diseases since they can automatically extract features of images.

The AI application starts with the preprocessing of the data that is a significant step in enhancing the performance of the model. During this phase, the images of crop leaves are made to a standard size to allow uniformity. The value of pixels are scaled to enhance the computational speed and data augmentation methods of rotation, flipping and scaling are implemented. The methods contribute to the enhancement of the diversity of datasets and decreasing overfitting.

The second step is training the model, in which the CNN model is developed using a labeled dataset sample of crop diseases. The data set has pictures of healthy plants and sick plants of various types. The model is trained on patterns like color changes, texture variations and disease specific patterns in leaves.

The CNN structure is composed of several layers, such as convolutional layers that extract features, pooling layers that reduce the number of dimensions, and fully connected layers that classify the features. This model is optimized through the use of minimizing prediction errors with the aid of optimization processes like gradient descent, which gradually increases its precision.

After the training, the model passes through the prediction stage, where it predicts new images that are uploaded by users. It provides the name of the disease, its confidence score, and the degree of the severity according to the learned patterns.

Lastly, the system also produces recommendations in accordance with the results of the prediction. These guidelines offer pesticides, fertilizers, and preventive measures as treatment options to curb the spread of diseases.

The AI model is implemented as an individual service and can interact with the backend via APIs which allows making predictions in real time.

3.7.4. System Integration

System integration refers to the process of integrating all the parts of the system to make a complete working application. In this initiative, the frontend, backend and AI model are integrated through a structured and linear workflow.

It starts by a user uploading a crop image via the frontend interface. A REST API request is sent to the backend with the image. The backend then sends the image to the AI model to do processing. The AI model performs an analysis of the image with the CNN algorithm and provides prediction results.

The results are then relayed back to the back-end, where they are added to the MongoDB database to be used later. The backend will then relay the results to the frontend where it gets presented to the user in a clear and organized format.

Such integration creates a smooth flow of data among system layers. It also keeps the separation of concerns in that each component is doing its own job but still they are a complete system.

Scalability is also achieved with the integration as each module can be enhanced or changed without impacting the whole system. To give an example, the frontend and backend are not required to be changed substantially to retrain or replace the AI model.

Generally, system integration guarantees effective communication, real-time processing and smooth user experience.

3.7.5. Summary of Implementation

The success of the AI-Powered Crop Disease Detection and Recommendation System is based on a successful integration of the contemporary web technologies and machine learning methods into one single solution. The user interface is intuitive and the frontend gives the user control, the backend handles system logic and data flow, and the AI model provides intelligent disease detection.

The modular architecture will be able to provide flexibility, scalability and maintainability of the system. All the components work autonomously but interact effectively via APIs. Such organization enables subsequent upgrades like adding additional crop data sets, enhancing model performance, or incorporating cutting-edge farm analytics.

All in all, the system is an effective, real-time, and user-friendly solution to crop disease detection, which can contribute greatly to agricultural decision-making and productivity.

4. Testing and Analysis

4.1. Test Plan

The test plan outlines the general approach to assessing the performance, accuracy, reliability and functionality of the AI-Powered Crop Disease Detection System. Testing is a critical process in software development since it helps in verifying whether the system can perform well as per the requirements and report the proper results to the users.

In this project, testing is primarily aimed at ensuring that the system has the ability to identify crop diseases using the images, process user inputs effectively and give useful recommendations without failure or crashing of the system.

To do this, the testing process will be split into two important levels:

- Unit Testing
- System Testing

Unit testing is done to test the individual components individually whereas system testing is done to test the complete integrated system. Both testing stages are significant to make sure that every component of the system will be operating properly and that all elements will be operating in harmony.

The test plan also assures that:

- I. The AI model produces accurate predictions
- II. The web application reacts appropriately to the user actions.
- III. It takes care of errors appropriately.
- VI. The system works effectively in various circumstances.

4.1.1. Unit Testing Plan

Unit testing refers to testing of single components or modules of the system in isolation. Procedures and processes are carried out in a manner that each module is tested individually and then integrated with the other components of the system. Unit testing is primarily to detect mistakes as early as possible and make sure that all functions are functioning as intended.

The following modules are tested on an individual basis in this project:

1. Image Upload Module

This module enables users to post pictures of crops. Testing will be done to make sure that the system does accept valid image formats (JPG, PNG) and not invalid inputs (text files or unsupported formats).

2. CNN Model (AI Prediction Module).

This is the essential module of the system. It takes the uploaded images and predicts crop diseases based on the trained CNN model. It is necessary to test the model because it is necessary to check that it is able to identify the diseases correctly and provide the correct results.

3. Result Display Module

This module shows the output of the prediction to the user. It is checked to make sure that the disease name, confidence score, and recommendations are displayed clearly and properly.

Unit Testing Objective.

The main objectives of unit testing are:

- To test the modules individually.
- To identify bugs at the initial stage of development.
- In order to achieve proper input-output behavior.
- In order to enhance the reliability of the systems prior to integration.

These modules are then tested by having predefined inputs and compare the obtained results with the expected results to ensure accuracy.

4.1.2. System Testing Plan

System testing is the testing of the entire system when all modules are integrated. It guarantees the functionality of the application as a unified entity and satisfies all the functional requirements.

In system testing, real-life situations are modeled to test system behavior in various conditions like valid input, invalid input and multiple user requests.

Table 3: Table of system testing plan

Test Scenario ID	Scenario Description	Expected Output	Actual Output	Result
ST01	Full system execution	Accurate prediction	Correct prediction shown	Pass
ST02	Invalid image input	Error message displayed	Error displayed correctly	Pass
ST03	Multiple image uploads	Stable performance	System handled smoothly	Pass
ST04	AI processing delay check	Fast response	Acceptable response time	Pass

Analysis of System Testing

System testing outcome indicates that the application is effective in all the situations tested. The system can effectively deal with the invalid and valid inputs provided without crashing or giving out wrong outputs. The AI model is well-integrated with the web app, and the general workflow works properly.

The system has also a response time that is within the acceptable limits thus confirming that the system is fit to be used in real-time with the agricultural environment.

4.2. Unit Testing

Unit testing results show how each module performs when tested individually.

Table 4: Table of the unit testing

Test Case ID	Module Tested	Input Type	Expected Output	Actual Output	Result
UT01	Image Upload	Valid Image	Image uploaded successfully	Image uploaded successfully	Pass
UT02	Prediction Model	Leaf Image	Disease name detected	Correct disease detected	Pass
UT03	Invalid Input	Text/File	Error message displayed	Error message displayed	Pass
UT04	Output Display	Prediction Data	Result shown clearly	Result shown correctly	Pass

Analysis of Unit Testing

The results of unit testing show that all the individual modules are operating properly. Testing did not reveal any significant errors or failures. All modules produced desired output using the input provided. This establishes that the internal elements of the system are stable, reliable, and in readiness to be integrated.

Unit testing also assisted in detecting small problems at an early stage which were rectified prior to integration of the system. This enhanced the overall performance and quality of the application.

4.3. System Testing

System testing is conducted once all the individual modules have been integrated into an entire working system. System testing tests the whole flow of the application as opposed to unit testing, which targets individual components.

The system is also tested in this stage to make sure that every part of the system functions well.

System Workflow Tested:

- User uploads an image of crop leaf.
- The system receives the image through the frontend
- The back-end processes and feeds the picture to the AI model.
- The disease is predicted via CNN model.
- The system will give the results back to the user interface.

Recommendations are displayed

- System Testing Objective.
- To have adequate module integration.
- To test the end-to-end system functionality.
- To test the stability of the system in actual use conditions.
- To ensure proper communication between frontend, backend and AI model.

System testing will assure that the complete application is a complete and integrated system.

4.4. Critical Analysis

The critical analysis will assess the overall performance, strengths, and limitations of the AI-Powered Crop Disease Detection System by the results of the testing.

- System Strengths

The system has a number of strengths. One is that it is extremely accurate in the detection of the disease because of the application of the Convolutional Neural Networks (CNN) which are very effective in the classification of images. Second, the system provides rapid processing and response time thus it is applicable in real-time. Third, the user-friendly interface makes the system user-friendly to even non-technical users like farmers. Finally, the system has reliable and clear recommendations, which assist the users to make informed agricultural decisions.

- System Limitations

Although the system has its strengths, it has certain limitations as well. The quality and size of the dataset is critical, as the quality of the AI model can decline with small or biased data. Also, the system might not work well with the unseen or rare forms of disease not covered in the training data. The system also needs the presence of internet and this might be restrictive in the remote regions. In addition, the quality of images posted by users can influence the performance, or have an unclear impact.

- Overall Evaluation

In general, the system has a massive contribution to the agricultural decision making through rapid and precise detection of crop diseases. It eliminates the need to rely on manual inspection and the availability of experts hence a viable and effective solution to farmers. By continuing to improve the system, e.g., by adding datasets, optimizing models, and providing support offline, it can be even more dependable and applicable to real-world agricultural settings.

5. Conclusion

This Final Year Project was able to achieve the main objective of developing and designing an AI-Powered Crop Disease Detection and Recommendation System to solve the most important problems in the agricultural industry. Agriculture is a major economic support especially in developing economies such as Nepal whereby a significant percentage of the population relies on agriculture as a source of livelihood. Nonetheless, conventional ways of observing crop diseases are usually time-consuming, erroneous, and extremely reliant on human judgment, which is not always easily available. This project is an adequate answer to these problems as it proposes a smart, robotic, and easy-to-use solution that makes use of innovations in Artificial Intelligence and web technologies.

Convolutional Neural Networks (CNNs) to detect diseases based on images were developed as a part of the system in this project, which is paired with a complete web application, created with the MERN stack (MongoDB, Express.js, React.js, and Node.js). The system can also be used to predict diseases by analyzing visual patterns, including discoloration, spots, and irregularities in the texture of the crops leaves by letting the user upload the images and the system can correctly identify the diseases. Besides detection, the system offers comprehensive and practical recommendations, such as organic treatment, the use of pesticides, and prevention. This makes sure that the platform is not only a diagnostic tool but also a full-fledged decision-support system to farmers.

The project was able to meet its set goals, such as creating a trained AI model, connecting it with a web-based interface, providing real-time predictive functionality, and storing previous data to analyze it. The application of features like user authentication, ability to upload images, delivery of recommendations and tracking of data was well realized, which shows the usefulness, usability and scalability of the system. The system can also be improved with the inclusion of historical data tracking that will allow users to track the trends in the disease and make in-depth long-term agricultural decisions.

The implementation of Scrum framework within the Agile approach was instrumental in the systematic and effective project development. This break-down of the process into cyclic sprints allowed the project team to constantly test, analyze and correct the AI model as well as the software modules. This iterative method was specifically useful in managing the dynamism of AI-based systems where performance of the models and system requirements change with time. Periodic

reviews and feedback systems were used to ensure that the project was on track and was able to meet the objectives and user requirements.

A number of challenges were experienced during the process of development. Availability and quality of the dataset to train the AI model were among the biggest challenges because the quality of the predictions largely depends on various and properly labeled data. Moreover, to optimize the model and achieve a higher accuracy this involved a lot of experimentation and tuning. Issues pertaining to technical challenges of integrating the AI model with the web application and facilitating the seamless communication between the frontend and backends were also discussed. Additionally, reliance on internet access was a constraint, particularly with the rural nature of the target population. The challenges were addressed in a logical manner using data preprocessing methods, model enhancement, effective API integration and design issues to enhance usability and performance.

Personal and professional development was also a significant achievement of this project. It increased technical capabilities in machine learning, deep learning, full-stack web development, and database management. In addition, it was a great learning experience on the implementation of software development methodologies, project schedule management, and real-life issues. The project strengthened the need to design solutions that are user-centered, particularly when solutions are to be developed to accommodate users who have different degrees of digital literacy.

In a wider sense, there are some social, economic and environmental impacts that are significant in the developed system. The system can promote early detection of crop diseases as farmers can take timely action and hence, minimize crop losses and promote higher output. The balanced treatment recommendations provided facilitate the responsible use of the pesticides and encourages sustainable agricultural practices, which help in conserving the environment. Moreover, the system will minimize the reliance on expensive expert consultations, and more advanced agricultural assistance will be affordable to small farmers.

Even though it has advantages, the system has some limitations. It relies on the quality and scope of the training data, and it might not do well with rare or never-before-seen diseases. Availability of internet connectivity to improve real-time predictions can also be a constraint to its application in remote regions where network infrastructure is poor. These restrictions indicate the possibilities to make improvements in the future, including the expansion of the dataset, better model generalization, and the integration of offline functionality.

To sum up, the current project has effectively indicated how AI-based technologies can be used to revolutionize the conventional agricultural systems, turning them into smart and data-driven ones. It is a practical and scalable solution that fills the gap between highly-technological solutions and

the practical needs of farming. The project does not only achieve its academic goals but also lays a solid base on what can be achieved in smart agriculture in future.

5.1. Legal, Social and Ethical Issues

The design and implementation of the AI-Powered Crop Disease Detection and Recommendation System are subject to a number of significant legal, social and ethical implications. Although the system has immense advantages in enhancing agricultural productivity and decisions, there are responsibilities that it brings on board and should be handled with care. These concerns make sure that the system is technically efficient as well as safe, fair, and acceptable to the real-life application. These issues will be tackled so as to gain confidence in users especially farmers and to maintain long term sustainability and responsible use of the technology.

5.1.1. Legal Issues

The legal aspects in this project are primarily associated with data protection, intellectual property and accountability of the system. As the system enables users to post crop pictures and can potentially hold user related information, data privacy and security will be a significant issue. Care should be taken to make sure that all user information is stored and cannot be accessed by unauthorized personnel. This may be through the application of encryption, secure log-in and access control measures. Lack of protection of user data can result in legal implications and loss of user confidence.

Intellectual property rights is another crucial legal factor. Data sets on which the Convolutional Neural Network (CNN) model is trained should be acquired in a legal manner. In case of the acquisition of images or datasets by external means, appropriate permissions or licenses should be taken. The unauthorized use of copyrighted materials may lead to legal arguments and fines. In the same way, the software created, both the AI model and web application, can also be safeguarded by the intellectual property rights, such as copyrights.

The legal issues that are of critical concern are also liability and responsibility. Given that the system offers disease detection and treatment suggestions, false prediction may cause loss of crops or loss of money among farmers. This begs the question of who bears the blame of such results. To counter this, the system must have disclaimers that the instructions are advisory and users must seek expert advice where they are required.

Also, the system should be in line with the information technology laws and regulations, particularly, the laws governing data protection and cybersecurity. The legal compliance also ensures protection of the users and enhances the credibility and reliability of the system.

5.1.2. Social Issues

Social problems are concerned with the impact of the system on people, communities, and society in general. The digital divide is one of the most significant issues as it means the difference between individuals, who have access to technology and those who do not. Farmers (and people in rural regions, in particular) might not have access to smartphones, internet, or the knowledge necessary to utilize these applications. This can restrict the usefulness and use of the system and may leave out the individuals who may require it the most.

Technology dependency is another key social challenge. Although the system yields correct and fast outcomes, it is possible that farmers will be over-reliant on it and start to lose their traditional knowledge and experience. Practical knowledge that has been handed down through generations has long been the foundation of agriculture, and technology might not necessarily be the best way to replace it.

The social impact of the system is also positive. It assists farmers to detect the diseases at an early stage, minimize crop losses and enhance productivity. It may also be useful especially where agricultural experts are not readily available. The system will enable farmers to make decisions on time because it offers immediate suggestions.

Social factors include language and usability. A lot of users might have local languages and some might be of low literacy. The system is therefore supposed to be made easy to comprehend, perhaps with some visual aids or multilingual facilities.

In general, although there are numerous social advantages of the system, one should make it inclusive, accessible and supportive but not substituting human knowledge wholesale.

5.1.3. Ethical Issues

Ethical problems are concerned with the moral obligations during the designing, developing, and utilization of the system. Accuracy and reliability is one of the most important ethical issues. Because farmers rely on the predictions of the system to make decisions, wrong predictions can be disastrous. Consequently, developers need to be careful that the AI model should be trained

with high quality and varied datasets and sustained with continuous improvement to ensure accuracy.

Transparency is another significant ethical factor. This is the point that users must be aware that the system is founded on Artificial Intelligence, and that predictions are not necessarily 100 per cent accurate. Calculating confidence scores and presenting results in a simple manner can assist the users to make better decisions and prevent misleading them.

Another important ethical issue is bias in AI models. In some instances, the model might not perform well in cases where the training dataset is not of a large diversity of crops or disease conditions. This may result in unjust or erroneous outcomes to certain users. Diversity in training data can be used to decrease bias and enhance fairness.

It is also critical to use user data ethically. The information gathered through users must not be utilized in the unauthorized commercial practices. Users need to know how their data is utilized and they need to control their information.

And finally, the problem of accountability. System providers and developers should be responsible in maintaining the system, correcting errors and enhancing performance. Ethical responsibility is what makes the system to remain useful to the users without harming them.

5.1.4. Conclusion of Section

To sum up, legal, social, and ethical issues are crucial to the effective deployment of the AI-Powered Crop Disease Detection and Recommendation System. By solving these problems, the system is guaranteed to be secure, fair, accessible and responsible. With these factors in mind, the system will be able to earn user confidence, increase its adoption, and make a positive contribution to the agricultural industry.

5.2. Advantages

The AI-Powered Crop Disease Detection and Recommendation System has a wide range of benefits enhancing the conventional farming methods. The system will offer an efficient, accurate and user-friendly solution to farmers and other agricultural stakeholders by combining Artificial Intelligence with current web technologies.

The high accuracy of the system in detecting diseases is one of the greatest benefits of the system. Convolutional Neural networks (CNN) allow the system to scan images of crops and diagnose diseases using the patterns it has learned (color variations, texture and shape). This decreases the human error that comes with manual inspection and enhances the dependability of illness diagnosis.

Time efficiency is another beneficial point. Common disease identification practices in the traditional aspect may involve farmers visiting professionals or waiting to have the disease analyzed in a laboratory, which may take a long time. Conversely, this system gives immediate or close to immediate results, which enables farmers to act on them immediately. Early diagnosis is important in preventing diseases and reducing losses to crops.

The system also improves the decision making process by not only identifying the disease but also offering treatment recommendations and preventive measures. This assists farmers to make a sound decision without having to depend solely on external specialists. Consequently, it minimizes the reliance of farming experts, particularly in remote locations where experts are inaccessible.

Another main benefit is the ease of access and usability by the users. It is user friendly and with simple interface the system is designed to suit users with very little technical knowledge. The characteristics of the system like simple picture uploading, well-defined results representation, and well-planned recommendations make it practical to be applied in real-life situations.

Also, the system enables storage of data and tracking of history, which can enable users to check previous predictions and track diseases with time. This assists in planning and management of crops in the long term.

Scalability and flexibility are guaranteed by the use of modern technologies, the MERN stack, and cloud-based infrastructure. The system is capable of supporting multiple users and can be easily upgraded with new features, better models or other crop datasets.

In general, the system is cost-effective, fast, and smart in managing crop diseases. It enhances the output in agriculture, minimizes crop losses and equips farmers with the latest technological equipment which is very favorable towards sustainable agricultural practices.

5.3. Limitations

Although the AI-Powered Crop Disease Detection and Recommendation System provides many benefits and has a great potential to be used in practice, it has a number of limitations influencing

the overall performance, usability, and applicability in real life. These limitations should be identified and comprehended to enhance the system and make it reliable in real agricultural setups.

The reliance on the quality and variety of training data is one of the most serious limitations of the system. Convolutional Neural Network (CNN) model is based on the previously trained data to perform predictions. In case the dataset lacks enough variations of types of crops, levels of disease, light conditions, or the environment, the model can provide biased or inaccurate outcomes. As an illustration, a disease that was captured in a particular light setting may not be identified when the input image is in a different setting. This renders the system less predictable with new or rare diseases which were not in the training set.

The reliance on the quality of the image is another significant constraint. The impression of the image is directly related to the accuracy of the system. The blurry, dark, partially obscured, or wrong angle images may result in false forecasts. In practical situations, farmers might not be able to get high quality cameras or may not be knowledgeable enough on taking appropriate images. Consequently, the quality of input can result in a poorly trained model giving incorrect results.

Internet dependency and access are also problematic in the system. As the application is based on cloud-based processing and communication of frontend, backend, and AI services, a stable internet connection is needed. Nevertheless, there is a lack of or poor internet access in many rural agricultural regions. This limits the functionality of the system by farmers who require it most hence limiting its practical use.

The other weakness is that it is not flexible to new and emerging diseases. Diseases of crops can be mutated or new diseases can develop with time. The AI model fails to adjust to such changes automatically and needs to be retrained periodically with new datasets. Unless updated periodically, the system can become old-fashioned and less efficient in detecting the current disease conditions.

Over-reliance on automated systems is also a problem. Although the system offers fast and helpful suggestions, users can rely wholly on the system and neglect the conventional farming knowledge or professional guidance. This may prove to be problematic when the system gives the wrong or incomplete recommendations, which may result in crop damage or loss of money.

Moreover, the system offers generalized solutions as opposed to highly customized solutions. It fails to take into account region specific factors like soil type, weather conditions, and local farming practices. Consequently, the treatments that are recommended are not necessarily the most effective to a particular location.

Lastly, there are security and privacy issues to be taken into account. As the system provides the storage of user data and images, the risk of unauthorized access or data misuse is present unless the proper security measures were introduced. User information must be put under protection to guarantee trust and ethical usage of the system.

5.3.1. Conclusion of Limitations

In summarization, the system offers an effective and innovative approach in detecting crop diseases, yet it has its challenges. Overcoming these constraints by providing superior datasets, offline support, more user instructions, updating the model on-the-fly and increasing the security level will greatly enhance the effectiveness and reliability of the system in practical use.

5.4. Future Work

Despite the fact that the modern stage of the project has already managed to create a working system base and resulted in the integration of AI, there are a number of significant aspects that can be developed in the future. They need to be improved to achieve better system intelligence, usability, scalability, reliability, and applicability to the real world. The next steps of work are the continuation of the improvement of the AI model, reinforcing the data analytics, and user engagement, as well as the further improvement of automation and system performance and reaching the status of full deployment. In sum, all these developments are intended to convert the system into an efficient prototype to a robust and production-oriented solution.

5.4.1. Elaborated Model Training and Lifelong Learning.

The first possible direction of work in the future is the continued training and optimisation of the AI model. Though the given model gives the first predictions with the acceptable level of accuracy, it can be improved considerably with the exposure to bigger and more different data. Heavy methods of model training like hyperparameter optimisation, ensemble learning, and transfer learning can be implemented to enhance prediction accuracy and generalisation (Goodfellow, Bengio and Courville, 2016).

The ability to include data that will be obtained in the real world in the process of using the system will allow learning continuously and making improvements. Since regions and seasons have different agricultural conditions, disease distribution, and other environmental factors, retraining on a regular basis will enable the model to be relevant and true with time. The iterative learning

technique has been highly suggested to artificial intelligence-based decision-support systems to guarantee the effectiveness in the long-term (Mitchell, 1997).

The further analysis of the model will also incorporate the advanced performance measurements, e.g. precision, recall, F1-score, receiver operating characteristic (ROC) curves. These measures give more insight into the performance of classification in addition to the simple accuracy, especially in imbalanced datasets which are typical in agricultural disease detection (Provost and Fawcett, 2013).

5.4.2. Improved Analytics Dashboards and Data Visualisation.

The other significant direction of work in the future is the creation of all-encompassing analytics dashboards. These dashboards will inform the stakeholders with valuable data based on the system information such as trends in the occurrence of diseases, accuracy of prediction, and user activities, and the performance of the system. Simplification, intuitive interpretation of complex data may be supported with the help of visual analytics tools, including bar charts, line graphs, heatmaps, and geographical mappings (Few, 2013).

To administrators, analytics dashboards will facilitate successful monitoring of the system, evaluation of its performance, and making decisions that are based on data. Trend analysis can assist with early detection of disease outbreaks and seasonal trends in the case of agricultural experts and researchers. Such are the data-oriented insights that add superior practical value of the system as they convert raw data into usable knowledge (Han, Kamber and Pei, 2012).

5.4.3. Notification systems and Intelligent Alert Systems.

The combination of intelligent notification and alert mechanisms is also planned in the future work. These systems will automatically notify the user about the crucial events like disease diagnosis responses, treatment prescriptions, unusual levels of risk, or updates of the system. The notifications can be sent in various formats, such as email, text message, or even in-app notifications, which will make them accessible and timely.

The Early warning systems are exceptionally useful in farming areas where there is a significant crop loss reduction when early intervention is performed. Proactive decision-making can also be facilitated by intelligent alerts with their level of prediction confidence or environmental threshold, and help the user develop trust in the system (Sommerville, 2016). The option to customise notification settings will also enhance user experience, as it will enable a person to regulate the nature and volume of received alerts.

5.4.4. Local, Teamwork, and Sharing of Knowledge.

The next step in the evolution of work is the addition of community and collaboration capabilities to the system to encourage the involvement and intention to use it by the user. These can be discussion groups, professional consultation courses, question and answer sections and peer-to-peer interaction areas. These capabilities offer users an incentive to exchange experiences, consult with professional advice and communicating best practices in respect to crop management and disease prevention.

Community-based systems improve both collaborative learning and building trust because they integrate AI-based recommendations with expert knowledge (Wenger, 1998). The measurement of moderation and the validation of specialists will be crucial to guarantee the accuracy of information and eliminate the dissemination of fake news. Through facilitating collaboration, the platform is capable of changing into a holistic knowledge ecosystem, instead of being a single diagnostic tool.

5.4.5. Scalability, Optimisation of performance, and Infrastructure Enhancement.

With the increased use of a system, the work in the future should be done in the context of scalability and performance optimisation. These comprise enhancing the backend architecture, optimising the database queries, increasing the efficiency of APIs and caching to minimize response time. There will be the need of load balancing and horizontal scaling policies to help mitigate the augmented user traffic without damaging system reliability (Bass, Clements and Kazman, 2013).

The implementation of cloud-native infrastructure and containerisation technology could also contribute to the increased scalability and flexibility of deployment. These methods allow being very resource management effective, fault tolerant and able to scale cost-effectively, which is essential in a real-world system deployment (Buyya et al., 2019).

5.4.6. Security, Privacy, and Regulatory Compliance.

Securing the systems and privacy of data will also be the priority of the future improvements. Since the platform will deal with user-generated and potentially sensitive farming information, it is important that strong security protocols are in place. These are secure authentication system, role based access control, encryptions of data and periodical security audits.

It is important to ensure that the appropriate data protection regulations are observed so as to ensure that user confidence in the system and credibility of the system is preserved. The principles of privacy-by-design must be implemented in further development in order to reduce the risks of data exposure and guarantee the ethical treatment of data (ISO/IEC, 2011). Enhancing security will also ensure that the system is secured against unauthorised access, data breach, and cyber threats.

5.4.7. Complete Implementation, testing and verification.

The last step in future employment is deployment of the entire system and large-scale real-life testing. This is comprised of rolling out the system into a production server, pilot testing on real users and getting systematic feedback that can be used to improve further. The real operating conditions will contribute to the evaluation of the system usability, reliability, and effectiveness, and the user acceptance testing and monitoring of the system will assist in that (Pressman and Maxim, 2020).

The evaluation of any system after deployment will be significant in detecting the limitations and tweaking the system features. An iterative improvement will be promoted with the help of continuous feedback loops, which will guarantee the system will develop as user requirements and environmental conditions are changing.

5.4.8. Conclusion of Future Improvements

Finally, the AI-Powered Crop Disease Detection and Recommendation System has a great potential to develop and advance further to become more efficient, accessible, and applicable to the real-life agricultural setting. Even though the existing version of the system shows a good level of disease detection and powerful recommendations, the future improvements can raise the work and usage of the system considerably.

It is one of the most significant enhancements the training dataset should be expanded and diversified. With the additions of more types of crops, variations of diseases, and environmental factors, it is possible to significantly increase the accuracy and generalization capability of the CNN model. This will assist the system to work better in the field of farming where there is a high level of variability.

The other major improvement is the addition of offline functionality, which would enable farmers in rural or remote locations with weak internet connectivity to still use it. This would render the application more accommodating and reachable. Besides, the incorporation of the real-time

environmental and weather information would also enhance the quality of recommendations as they would be more context-oriented and location-specific.

Further evolution of a mobile version of the application would also enhance usability since farmers would be able to take and analyze crop pictures on their smartphones. The inclusion of more languages would also make it more accessible to the users who may have other linguistic backgrounds.

Furthermore, the ongoing updates of models and learning processes can make the system up to date with new diseases and agricultural conditions. Large-scale and automated crop surveillance can also be investigated in conjunction with the integration with the IoT devices and the use of drone-based surveillance.

All in all, the following improvements will make the system smarter, more scalable, and user-friendly to farmers, which will play a vital role in sustainable agriculture and better crop productivity.

6. References

The references list is very essential in sustaining the quality and credibility of this report. All materials used and cited during the study are cited using the Harvard Anglia referencing style that is commonly used by UK institutions of higher learning. This ensures that there is uniformity in referencing format, and readers find it easy to trace and confirm the sources of original work.

References used are a mix of scholarly and professional literature i.e. textbooks, peer-reviewed journal articles, conference papers, industry reports and respectable online sources. All of the sources have been critically chosen to reinforce theoretical underpinnings, methodological, system design, and analysis used in the report. There is an in-text citation in which the original authors of ideas, models, frameworks, and technical concepts that are being discussed are mentioned.

The use of the Harvard Anglia style also builds the academic integrity as all the borrowed ideas, quotes, and paraphrased materials are referenced. The practice prevents plagiarism and can be considered good research behaviour. Moreover, proper referencing supports validity of the report because it indicates that the conclusions and design choices are based on existing research and best practices in software engineering, artificial intelligence as well as project management fields.

On the whole, the references list gives a clear account of intellectual sources that were used to develop the project and justify the confidence and intellectual quality of the work.

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7. Bibliography

The bibliography is the list of all academic, technical and web sources used and referred to in the development of this project. It contains academic books, peer-reviewed journal publications, research papers, and official documents that helped in the theoretical background, system design, and development of the AI-Powered Crop Disease Detection and Recommendation System.

The main idea of bibliography is to maintain academic integrity by making a proper recognition of the original authors and information sources. It shows that the work is based on quality and authoritative research; thus, preventing plagiarism and maintaining ethical academic principles. Also, it offers transparency as it enables other readers and evaluators to follow the origin of the ideas, confirm the information provided, and pursue the topic area more.

Within the framework of this project, the bibliography reinforces such technical and theoretical elements as Artificial Intelligence, Deep Learning, Convolutional Neural Networks (CNNs), Agile and Scrum methodologies, and full-stack web development based on the MERN stack. The project references the existing literature and industry-standard documentation, thus correlating its implementation with the existing practices and technological progress.

Besides, the presence of a properly designed bibliography increases the credibility, reliability, and scholarly level of the report. It reflects the extent of research done and denotes a critical involvement with the current knowledge in the field. It is also a valid source of future researchers, developers and practitioners who might want to further develop, reproduce or even enhance this work.

To conclude, a bibliography is a fundamental part of this report which helps not only in ensuring academic honesty but also in the strength, validity, and scholarly value of the project.

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8. Appendix

The supplementary material presented in the appendix does not interfere with the flow of the main body of the report. It also consists of meticulous evidence of the process of creation, technical artefacts, and documentation created during project lifecycle. Such materials have been added in order to make it more transparent, show the practical implementation and give verifiable evidence about the progress and result attained.

Iteration documents and evidences are documents that are typically acquired through iterative processes.

8.1. Iteration Documents and Evidence

These are documents that are usually obtained using iterative processes.

This subsection contains full supporting material which shows the development and evaluation that was done in the project in an iterative manner. The appendix presents verifiable statements on planning, implementation, and review actions performed in several cycles of implementation, which is an expression of agile development approach.

The documentation contains sprint planning and sprint review documents, and they contain iteration objectives, allocated tasks, and anticipated deliverables. These reports show the way project requirements were converted into development goals to be implemented systematically as well as how such an evaluation at the end of every iteration was carried out. Iteration logs also capture the development of what can be seen as improvements carried out with regard to technical issues, stakeholder feedback, and changing project requirements.

Furthermore, pre-survey tools and feedbacks are also given to show how preliminary user needs and expectations were determined and confirmed. The survey tools such as structured questionnaires were aimed at collecting information about user requirements, the challenges, and features preferences to the proposed system. The survey responses to the same provide empirical data to substantiate major design and functionality changes, which ought to be made at the initial stages of development. The fact that both the instruments and their responses are included makes the system transparent and it helps to make sure that the system design was based on user-centred research and not assumptions.

Other materials produced in the development process like configuration files, testing evidence, validation output, and iteration summaries are added where necessary. The sum of the iteration documents and supporting evidence gives a full audit trail of the lifecycle of project development,

which allows the evaluators assess the development process and the results of the project development in a systematic and transparent way.

8.2. Pre- Survey

8.2.1. Pre-Survey Overview

The AI-Powered Crop Disease Detection and Recommendation System was developed and implemented after a pre-survey was carried out. The general objective of the survey was to know the current practices, problem, and level of awareness of users about crop disease detection techniques. The survey was conducted among 30 respondents who were either farmers, students, agricultural experts or other professionals. The survey contributed to the realization of the shortcomings of the conventional methods of detecting diseases and offered a strong rationale of the necessity of an AI-based solution.

8.2.2. Pre-Survey Results Overview.

Pre-survey findings indicate that most of the respondents were students who constituted approximately 76.7% of the total participants with the remaining respondents being farmers, traders and other professionals. This shows that the survey was conducted on a composite group of users and offers a wider range of view of the agricultural awareness and use of technology.

Regarding experience in crop disease, the majority of the users said that they only have crop disease issues sometimes or rarely, with 46.7% of the respondents saying they have crop disease issues sometimes and 36.7% saying sometimes. A lesser percentage had frequent or no disease problems. This implies that the occurrence of crop disease is prevalent but not uniform among respondents.

In terms of the present disease detection tools, 40 percent of the users make use of visual observation and 56.7 percent of users make use of agricultural experts. Also 46.7% refer to internet resources like YouTube as a guide. Only a low percentage (6.7%), indicated that they had no appropriate way of identifying disease. This underlines the fact that majority of the users are still relying on manual or external means of support as opposed to automated systems.

In response to the question on the importance of early disease detection, 53.3 percent of the respondents affirmed that it is important, 36.7 percent not sure or maybe and 10 percent not sure.

This indicates that the knowledge on early disease detection is moderate and not well-developed among users.

The survey also showed that the users are moderately familiar with mobile applications with 56.7 percent being somewhat familiar, 16.7 percent familiar and 26.7 percent being not familiar with mobile apps. This is a demonstration of different degrees of digital literacy of participants.

Some of the challenges that were reported by users when determining the diseases affecting crops manually were lack of knowledge, inability to recognize the symptoms, and a combination of several problems. This confirms that conventional practices are not necessarily dependable and convenient.

Regardless of these obstacles, an interest in a solution built on artificial intelligence was high. Many of the users were willing to use a mobile application to detect diseases and the majority of users emphasized the fact that accuracy is crucial in these systems. The most popular device identified to access such applications was smartphones.

8.2.3. End of Pre-Survey.

To sum it up, the pre-survey outcomes clearly show that the majority of users use conventional techniques like visual inspection and expert consultation to detect crop diseases. These techniques however are usually constrained by accuracy, accessibility and time. Meanwhile, the interest of the users in an AI-based solution was high, particularly, in an accurate, user-friendly, and smartphone-based solution. The results are a solid argument to develop the AI-Powered Crop Disease Detection and Recommendation System as one of the modern tools to address the current problems in agriculture.

Pre- Survey Form Link: <https://forms.gle/6o3sQLGfRLePynt78>

Pre-Survey Questionnaire (AI Crop Disease Detection System)

This research is part of the Final Year Project titled "AI Crop Disease Detection System". Your feedback and responses will play an important role in guiding the development and improvement of the project. Your time and effort are sincerely appreciated. Thank you.

1. Email: neeska2006@gmail.com

2. YOUR NAME: Neeska Neelane

3. 1. What is your occupation?

Mark only one oval.

☐ a) Farmer

☒ b) Farmer

☐ c) Agricultural expert

☐ d) Trader Business

☐ e) Other: _____

4. 3. How often do you face crop disease problems?

Mark only one oval.

☐ a) In person

☒ b) In person

☐ c) In person

☐ d) In person

☐ e) Other: _____

5. 3. How do you currently identify crop diseases?

Mark only one oval.

☐ a) Visual observation

☐ b) Asking an expert

☒ c) Using a mobile app

☐ d) No answer

☐ e) Other: _____

6. 4. Do you think early detection of disease can reduce crop loss?

Mark only one oval.

☐ a) Yes

☐ b) No

☒ c) Maybe

☐ d) Not sure

☐ e) Other: _____

7. 5. How familiar are you with using mobile apps for agriculture?

Mark only one oval.

☐ a) Very familiar

☒ b) Somewhat familiar

☐ c) Not familiar

☐ d) I don't use mobile apps

8. 6. What challenges do you face when identifying plant diseases manually?

Mark only one oval.

☐ a) Lack of knowledge

☐ b) Hard to recognize symptoms

☒ c) All of the above

9. 7. Would you use a mobile app that detects crop diseases using photos?

Mark only one oval.

☒ a) Definitely yes

☐ b) Maybe

☐ c) No

☐ d) Not sure

10. 8. How important is accuracy in a disease detection app?

Mark only one oval.

☐ a) Extremely important

☒ b) Important

☐ c) Moderate important

☐ d) Not important

11. 9. What type of crops do you grow the most?

Check all that apply.

☐ a) Rice

☐ b) Wheat

☒ c) Sugarcane

☐ d) Potato

☐ e) All of the above

12. 10. Would you prefer an app that also gives treatment suggestions?

Mark only one oval.

☒ a) Yes, essential

☐ b) Yes, optional

☐ c) Not necessary

☐ d) No

13. 11. What device would you mostly use for disease detection?

Mark only one oval.

☒ a) Smartphone

☐ b) Tablet

☐ c) Laptop

☐ d) None

14. 12. How confident are you in taking clear photos of crops?

Mark only one oval.

1 2 3 4 5

☐ a) 1

☐ b) 2

☐ c) 3

☒ d) 4

☐ e) 5

15. 13. Which features would you find most useful in the app?

Mark only one oval.

☐ a) Disease identification

☒ b) Treatment suggestions

☐ c) Forecasting crop loss

☐ d) Weather-related crop alerts

☐ e) All of the above

16. 14. Do you prefer the app to work offline?

Mark only one oval.

☐ a) Yes

☐ b) No

☒ c) Doesn't matter

17. 15. How likely are you to recommend such an AI app to others?

Mark only one oval.

☐ a) Very likely

☒ b) Likely

☐ c) Fairly likely

☐ d) Not likely

18. 16. What is your major concern about AI in agriculture?

Check all that apply.

☐ a) Accuracy

☒ b) Cost

☐ c) Data security

☒ d) Reliability of data

☐ e) No concerns

19. 17. Do you believe AI can help increase agricultural productivity?

Mark only one oval.

☐ a) Yes

☐ b) No

☒ c) Maybe

20. 18. How satisfied are you with current disease detection methods?

Mark only one oval.

1 2 3 4 5 6 7 8 9 10

☐ a) 1

☐ b) 2

☐ c) 3

☐ d) 4

☒ e) 5

☐ f) 6

☐ g) 7

☐ h) 8

☐ i) 9

☐ j) 10

21. 19. Would you pay a small fee for accurate disease detection?

Mark only one oval.

☐ a) Yes

☒ b) Maybe

☐ c) No

22. 20. What language would you prefer the app to use?

Mark only one oval.

☐ a) English

☒ b) Nepali

☐ c) Local language

☐ d) Other: _____

23. "Please feel free to leave any feedback, comments, or suggestions for improvement. Your thoughts will be very helpful in enhancing our AI Crop Disease Detection System."

Yes all good.

The content is neither product nor endorsed by Google

Google Forms

Figure 10: Image of a Pre-survey form 1

Pre-Survey Questionnaire (AI Crop Disease Detection System)

The survey is part of the Final Year Project titled "AI-Based Crop Disease Detection System". Your feedback and responses will play an important role in guiding the development and improvement of this project. Your time and effort are sincerely appreciated. Thank you.

1. Email: Achyut.Paraajuli@gmail.com

2. YOUR NAME: Achyut Paraajuli

3. 1. What is your occupation?

Mark only one oval.

☐ a) Farmer

☒ b) Student

☐ c) Agricultural expert

☐ d) Teacher/Researcher

☐ Other: _____

4. 2. How often do you face crop disease problems?

Mark only one oval.

☒ a) Frequently

☐ b) Sometimes

☐ c) Rarely

☐ d) Never

☐ Other: _____

5. 3. How do you currently identify crop diseases?

Mark only one oval.

☐ a) Visual observation

☒ b) Asking an expert

☐ c) Smartphone App/Tool

☐ d) No method

☐ Other: _____

6. 4. Do you think early detection of disease can reduce crop loss?

Mark only one oval.

☐ a) Yes

☐ b) No

☒ c) Maybe

☐ d) Not sure

☐ Other: _____

7. 5. How familiar are you with using mobile apps for agriculture?

Mark only one oval.

☐ a) Very familiar

☒ b) Somewhat familiar

☐ c) Not familiar

☐ d) I don't use mobile apps

8. 6. What challenges do you face when identifying plant diseases manually?

Mark only one oval.

☐ a) Lack of knowledge

☐ b) Hard to recognize symptoms

☒ c) All of the above

9. 7. Would you use a mobile app that detects crop diseases using photos?

Mark only one oval.

☐ a) Definitely yes

☒ b) Maybe

☐ c) No

☐ d) Not sure

10. 8. How important is accuracy in a disease detection app?

Mark only one oval.

☐ a) Extremely important

☐ b) Important

☒ c) Somewhat important

☐ d) Not important

11. 9. What type of crops do you grow the most?

Check all that apply.

☐ a) Rice

☐ b) Wheat

☒ c) Vegetables

☒ d) Fruits

☐ e) All of the above

12. 10. Would you prefer an app that gives treatment suggestions?

Mark only one oval.

☒ a) Yes, essential

☐ b) Yes, optional

☐ c) Not necessary

☐ d) No

13. 11. What device would you mostly use for disease detection?

Mark only one oval.

☒ a) Smartphone

☐ b) Tablet

☐ c) Laptop

☐ d) None

14. 12. How confident are you in taking clear photos of crops?

Mark only one oval.

1 2 3 4 5

Never ☐ ☐ ☒ ☐ ☐ Always

15. 13. Which features would you find most useful in the app?

Mark only one oval.

☐ a) Disease identification

☒ b) Treatment suggestions

☐ c) Prevention tips

☐ d) Weather-related crop tips

☐ e) All of the above

16. 14. Do you prefer the app to work offline?

Mark only one oval.

☐ a) Yes

☐ b) No

☒ c) Doesn't matter

17. 15. How likely are you to recommend such an AI app to others?

Mark only one oval.

☐ a) Very likely

☒ b) Likely

☐ c) Unlikely

☐ d) Not sure

18. 16. What is your major concern about AI in agriculture?

Check all that apply.

☒ a) Accuracy

☒ b) Cost

☒ c) Trust issues

☐ d) Difficulty of use

☐ e) No concerns

19. 17. Do you believe AI can help increase agricultural productivity?

Mark only one oval.

☐ a) Yes

☐ b) No

☒ c) Maybe

20. 18. How satisfied are you with current disease detection methods?

Mark only one oval.

1 2 3 4 5 6 7 8 9 10

Not at all ☐ ☐ ☐ ☐ ☒ ☐ ☐ ☐ ☐ ☐ Very satisfied

21. 19. Would you pay a small fee for accurate disease detection?

Mark only one oval.

☐ a) Yes

☒ b) Maybe

☐ c) No

22. 20. What language would you prefer the app to use?

Mark only one oval.

☐ a) English

☒ b) Nepali

☐ c) Local language

☐ d) Other: _____

23. "Please feel free to leave any feedback, comments, or suggestions for improvement. Your thoughts will be very helpful in enhancing our AI Crop Disease Detection System."

It should be efficient to use.

The content is neither made nor endorsed by Google

Google Forms

Figure 11: Image of Pre-Survey Form 2

Pre-Survey Questionnaire (AI Crop Disease Detection System)

This questionnaire is part of the Pre-Survey Questionnaire of the AI Crop Disease Detection System. Your feedback and responses will play an important role in guiding the development and improvement of the project. Your time and input are greatly appreciated. Thank you.

1. Email: neseka2006@gmail.com

2. YOUR NAME: Neeseka Neupane

3. 1. What is your occupation?

Mark only one oval.

☒ a) Farmer
☐ b) Researcher
☐ c) Agricultural expert
☐ d) Teacher/Trainer
☐ e) Other _____

4. 2. How often do you face crop disease problems?

Mark only one oval.

☐ a) Frequently
☒ b) Sometimes
☐ c) Rarely
☐ d) Never
☐ e) Other _____

5. 3. How do you currently identify crop diseases?

Check all that apply.

☐ a) Visual observation
☐ b) Asking an expert
☒ c) Internet/YouTube
☐ d) No method
☐ e) Other _____

6. 4. Do you think early detection of disease can reduce crop loss?

Mark only one oval.

☐ a) Yes
☐ b) No
☒ c) Maybe
☐ d) Not sure
☐ e) Other _____

7. 5. How familiar are you with using mobile apps for agriculture?

Mark only one oval.

☐ a) Very familiar
☒ b) Somewhat familiar
☐ c) Not familiar
☐ d) I don't use mobile apps

8. 6. What challenges do you face when identifying plant diseases manually?

Mark only one oval.

☐ a) Lack of knowledge
☐ b) Hard to recognize symptoms
☒ c) All of the above

9. 7. Would you use a mobile app that detects crop diseases using photos?

Mark only one oval.

☒ a) Definitely yes
☐ b) Maybe
☐ c) No
☐ d) Not sure

10. 8. How important is accuracy in a disease detection app?

Mark only one oval.

☐ a) Extremely important
☒ b) Important
☐ c) Moderately important
☐ d) Not important

11. 9. What type of crops do you grow the most?

Check all that apply.

☐ a) Rice
☐ b) Wheat
☒ c) Maize
☐ d) Potato
☐ e) All of the above

12. 10. Would you prefer an app that also gives treatment suggestions?

Mark only one oval.

☒ a) Yes, essential
☐ b) Yes, optional
☐ c) No, not necessary
☐ d) No

13. 11. What device would you mostly use for disease detection?

Mark only one oval.

☒ a) Smartphone
☐ b) Tablet
☐ c) Laptop
☐ d) None

14. 12. How confident are you in taking clear photos of crops?

Mark only one oval.

1 2 3 4 5
☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5

15. 13. Which features would you find most useful in the app?

Mark only one oval.

☐ a) Disease identification
☒ b) Treatment suggestions
☐ c) Forecasting
☐ d) Weather-related crop alerts
☐ e) All of the above

16. 14. Do you prefer the app to work offline?

Mark only one oval.

☐ a) Yes
☐ b) No
☒ c) Doesn't matter

17. 15. How likely are you to recommend such an AI app to others?

Mark only one oval.

☐ a) Very likely
☒ b) Likely
☐ c) Unlikely
☐ d) Never

18. 16. What is your major concern about AI in agriculture?

Check all that apply.

☐ a) Accuracy
☒ b) Cost
☐ c) Data issues
☒ d) Difficulty of use
☐ e) No concerns

19. 17. Do you believe AI can help increase agricultural productivity?

Mark only one oval.

☐ a) Yes
☐ b) No
☒ c) Maybe

20. 18. How satisfied are you with current disease detection methods?

Mark only one oval.

1 2 3 4 5 6 7 8 9 10
 Satisfied ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☒ 9 ☐ 10

21. 19. Would you pay a small fee for accurate disease detection?

Mark only one oval.

☐ a) Yes
☒ b) Maybe
☐ c) No

22. 20. What language would you prefer the app to be in?

Mark only one oval.

☐ a) English
☒ b) Nepali
☐ c) Local language
☐ d) Other _____

23. "Please feel free to leave any feedback, comments, or suggestions for improvement. Your thoughts will be very helpful in enhancing our AI Crop Disease Detection System."

Yes all good.

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Google Forms

Figure 12: Image of Pre-Survey Form 3

Pre-Survey Questionnaire (AI Crop Disease Detection System)

The researcher is part of the Final Year Project titled AI Based Crop Disease Detection System. Your feedback and comments will play an important role in studying the development and improvement of this project. Your time and input are sincerely appreciated. Thank you.

1. Email: np05cx4230029@gmail.com

2. YOUR NAME: Pennu Bohra

3. What is your occupation?

Mark only one oval.

☐ a) Farmer ☒ b) Teacher ☐ c) Agricultural expert ☐ d) Teacher/Trainer ☐ Other: _____

4. How often do you face crop disease problems?

Mark only one oval.

☐ a) Frequently ☒ b) Sometimes ☐ c) Rarely ☐ d) Never ☐ Other: _____

5. How do you currently identify crop diseases?

Check all that apply.

☐ a) Visual observation ☒ b) Asking an expert ☐ c) Internet/YouTube ☐ d) No method ☐ Other: _____

6. Do you think early detection of disease can reduce crop loss?

Mark only one oval.

☒ a) Yes ☐ b) No ☐ c) Maybe ☐ d) Not sure ☐ Other: _____

7. How familiar are you with using mobile apps for agriculture?

Mark only one oval.

☐ a) Very familiar ☐ b) Somewhat familiar ☒ c) Not familiar at all ☐ d) Don't use mobile app

8. What challenges do you face when identifying plant diseases manually?

Mark only one oval.

☐ a) Lack of knowledge ☒ b) Hard to recognize symptoms ☐ c) All of the above

9. Would you use a mobile app that detects crop diseases using photos?

Mark only one oval.

☒ a) Definitely yes ☐ b) Maybe ☐ c) No ☐ d) Not sure

10. How important is accuracy in a disease detection app?

Mark only one oval.

☒ a) Extremely important ☐ b) Important ☐ c) Somewhat important ☐ d) Not important

11. What type of crops do you grow the most?

Check all that apply.

☐ a) Rice ☐ b) Wheat ☐ c) Vegetables ☐ d) Fruits ☒ All of the above

12. Would you prefer an app that also gives treatment suggestions?

Mark only one oval.

☐ a) Yes, essential ☒ b) Yes, optional ☐ c) Not necessary ☐ d) No

13. What device would you mostly use for disease detection?

Mark only one oval.

☒ a) Smartphone ☐ b) Tablet ☐ c) Laptop ☐ d) None

14. How confident are you in taking clear photos of crops?

Mark only one oval.

1 2 3 4 5

Confidence: ☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5

15. Which features would you find most useful in the app?

Mark only one oval.

☒ a) Disease identification ☐ b) Treatment suggestions ☐ c) Prevention tips ☐ d) Notifications and crop alerts ☐ All of the above

16. Do you prefer the app to work offline?

Mark only one oval.

☒ a) Yes ☐ b) No ☐ c) Doesn't matter

17. How likely are you to recommend such an AI app to others?

Mark only one oval.

☒ a) Very likely ☐ b) Likely ☐ c) Unlikely ☐ d) Not at all

18. What is your major concern about AI in agriculture?

Check all that apply.

☒ a) Accuracy ☒ b) Cost ☒ c) Training data ☒ d) Reliability of AI ☐ e) No concerns

19. Do you believe AI can help increase agricultural productivity?

Mark only one oval.

☒ a) Yes ☐ b) No ☐ c) Maybe

20. How satisfied are you with current disease detection methods?

Mark only one oval.

1 2 3 4 5 6 7 8 9 10

Satisfaction: ☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

21. Would you pay a small fee for accurate disease detection?

Mark only one oval.

☒ a) Yes ☐ b) Maybe ☐ c) No

22. What language would you prefer the app to use?

Mark only one oval.

☒ a) English ☐ b) Nepali ☐ c) Local language ☐ d) Other: _____

23. "Please feel free to leave any feedback, comments, or suggestions for improvement."

Your thoughts will be very helpful in enhancing our AI Crop Disease Detection System.

The content is neither created nor modified by Google

Google Forms

Figure 13: Image of Pre- Survey Form 4

8.3. Post- Survey

Post- Survey Form Link : <https://forms.gle/5DcGU5F2vE39ow6K8>

Post-implementation and testing of AI-Powered Crop Disease Detection and Recommendation System was followed by a post-survey involving 30 participants to assess the performance of the system, its usability, accuracy, and effectiveness in the real-life application of agriculture.

i. Overall User Experience

The findings depict a very positive user reaction.

- 42.1% of users evaluated the system as excellent.
- 42.1% rated it as “Good”
- 15.8% rated it as “Average”
- 0% rated it as being poor.

This shows clearly that most users (84.2) had a positive experience with the system which means that the system is able to deliver on what the users expect of them.

ii. System usability.

The system was also discovered to be easy to use and user-friendly:

- 57.9 percent of users responded that the system is Easy to use.
- 26.3% said it was “Very easy”
- 15.8 percent gave it a rating of “Moderate.
- 0% struggled with it.

This demonstrates that 84.2 percent of users considered the system easy or very easy to use, which proves that the interface is user-friendly and can be used by farmers with limited technical skills.

iii. Precision of Disease Diagnosis.

In terms of disease detection performance:

- Accuracy was rated as Accurate by 57.9% of the users.
- 31.6 percent found it as “Somewhat accurate.
- 10.5% gave it a score of Very accurate.
- 0% reported inaccurate results

This shows that even with practical conditions, the CNN model is very reliable because 100% of users believed that the predictions provided by the system were at least partially correct or higher.

iv. Usefulness of Recommendations

The suggestions and recommendations of the treatment were also much valued:

- 55.6% rated them as Very useful.
- 38.9% found them “Useful”
- 5.5% considered them as a bit useful.
- 0% considered them as Not useful.

This demonstrates that 94.5% of users found the recommendations useful or very useful, which confirms that the system can offer valuable agricultural assistance beyond detection.

v. Performance and Speed

System performance was also tested by the users:

- The percentage of those who claimed the system was Fast was 52.6%.
- 15.8% said “Very fast”
- 26.3% rated it as “Average”
- 5.3% said “Slow”

In general, 68.4% of users reported quick or very quick performance, which means that the system is responsive in the majority of cases.

vi. Reliability and Decision Support.

The effect of the system on the agricultural decisions was also positive:

- 52.6 stated that they would trust it (“Mostly yes).
- 36.8% were “Not sure”
- 10.5% had complete trust in the system.

This indicates that the majority of the users have confidence in the system, but it is possible to enhance the confidence by increasing the accuracy and awareness.

vii. User Satisfaction and Feedback.

General satisfaction with the interface of the system:

- 42.1% were “Satisfied”
- 21.1% were “Very satisfied”
- 36.8% were “Neutral”
- 0% were dissatisfied

This means that 63.2 percent of users are satisfied or highly satisfied but the UI is suggested to be improved.

viii. User Suggestions

The general user comments were:

- Enhance user interface design and layout.
- Minimise minor technical difficulties.
- Enhance precision of rare cases.
- Introduce additional functionality and offline capabilities.

8.3.1. Overall Conclusion

To sum up, the post-survey data on 30 users vividly proves that AI-Powered Crop Disease Detection System is efficient, trustworthy, and convenient to use. Most users provided positive feedback in terms of accuracy, usability, and usefulness. Having more than 80 percent positive scores in majority of the categories, the system is a viable solution to real-life agricultural issues. The system can still be made more efficient by minor changes to the design of the UI, optimizing its performance, and increasing the volume of data to which users can trust the system in the future.

8.4.2. Comparison between Pre and Post Survey (Analysis in Paragraph Form)

The comparison of the pre-survey and post-survey results provides a clear indication of the enhancement of the user awareness, usability, perception of accuracy, and overall satisfaction with the AI-Powered Crop Disease Detection and Recommendation System. The pre-survey was aimed at the comprehension of the current practices whereas the post-survey was aimed at the assessment of the implemented system.

During the pre-survey, majority of the respondents were very dependent on the traditional ways of diagnosing crop diseases. It was found that approximately 40 percent of them relied on visual inspection, 56.7 percent consulted with experts, and 46.7 percent visited internet resources, such as YouTube. It means that in the past, farmers and users relied on manual or external assistance which is slower and may not always be available. Nevertheless, during the post-survey, users reported that the system would save them a lot of reliance on experts, with 63.2% claiming significant decrease and 31.6% claiming that it decreased. This is a clear indication that the system substituted conventional dependency with an automated system of AI-based solutions.

In terms of ease of use, the pre-survey revealed moderate familiarity with mobile application, with only 16.7% of the respondents claiming to be very familiar with it, and more respondents reporting having limited experience with mobile application. However, the post-survey results demonstrate one significant improvement, as the share of users who found the system easy or very simple to

use is significantly higher (84.2% of users found the system easy or very easy to use), which indicates that the interface design was efficient and user-friendly even among non-technical users.

With regards to awareness and effectiveness of disease detection, the pre-survey revealed that a small percentage of users (53.3% of the respondents) felt that early detection would help lessen crop loss, and most users were unsure. Following the system usage, the post-survey found that 100 percent of the users accepted the system predictions as at least somewhat correct with 57.9 percent rating it correct and 10.5 percent very correct with a significant increase in trust and confidence in the AI-based detection.

In the case of decision-making support, the issues that were noted during the pre-survey were lack of knowledge and inability to recognize symptoms. Following implementation, 94.5 percent of users believed that they were useful or very useful in treatment and demonstrates that the system effectively turned raw detection into active agricultural advice.

Lastly, the general satisfaction went up. Although the pre-survey was indicative of uncertainty concerning AI-based farming tools, the post-survey indicated that, 84.2% of those who used the system ranked the system as good or excellent, and more than 80% said they would either use or recommend the system again. This shows a high positive user perception change on post system implementation.

Conclusion of Comparison

In general, the pre-survey and post-survey outcomes reveal that the AI-Powered Crop Disease Detection System has made a great contribution to user awareness, decreased reliance on manual tools, boosted confidence in disease detection, and improved decision-making. The system has been able to change the old ways to a more efficient technology-oriented system that is of great advantage to the present day agriculture.

8.4. Sample Codes / Development

8.4.1. Product Backlog

Product Backlog Product Backlog is the list of those features, requirements, additions, or fixes needed in a software system, in order and constantly changing. It is the sole provider of the requirements upon any alterations to be done to the product and is fundamental in the Scrum model. Product Backlog items are usually defined as user stories, a system functionality that is described through the eyes of end users or stakeholders (Schwaber and Sutherland, 2020).

Product Backlog is also prioritized in terms of business value, risk and dependency where the most important features are built at the beginning. Agile development does not have a fixed backlog, but the backlog is enhanced on a regular basis based on the changing user requirements, feedback, and project constraints. When the Sprint Backlog is being planned, items are taken out of the Product Backlog and into the Sprint Backlog, which allows it to be developed incrementally and iteratively.

The functional requirements that are reflected in the Product Backlog in this project include user authentication, AI-based crop disease detection, treatment recommendation, analytics, and administrative controls. The project is also well structured through its Product Backlog, which makes it have transparency, good planning, and adherence to the Agile and Scrum concepts during the development lifecycle.

Table 5: Table of Product Backlog

ID	User Story	Priority	Sprint	Story Point
US001	As a farmer, I want to register an account so that I can use the system securely.	High	1	3
US002	As a farmer, I want to log in using my credentials so that I can access my dashboard.	High	1	2
US0003	As a farmer, I want to upload an image of a crop leaf so that the system can detect diseases.	High	1	5
US004	As a system, I want to preprocess uploaded images so that AI predictions are accurate.	High	1	5
US005	As a system, I want to classify crop diseases using a CNN model.	High	2	8
US006	As a farmer, I want to view disease prediction results so that I know the crop condition.	High	2	3
US007	As a farmer, I want to see a confidence score with predictions so that I can trust results.	Medium	2	2
US008	As a farmer, I want to receive treatment recommendations so that I can cure the disease.	High	2	5
US009	As a farmer, I want organic treatment suggestions so that I can avoid chemical damage.	Medium	3	3
US010	As a farmer, I want pesticide recommendations with dosage so that treatment is safe.	High	3	5

US011	As a system, I want to store prediction history so that farmers can track past results.	High	3	5
US012	As a farmer, I want to view my prediction history so that I can monitor crop health.	Medium	3	3
US013	As a farmer, I want disease trend charts so that I can identify frequent issues.	Medium	4	3
US014	As a system, I want to generate analytics reports so that trends are visualized clearly.	Medium	4	5
US015	As an admin, I want to manage crop and disease datasets so that AI accuracy improves.	High	4	8
US016	As an admin, I want to retrain AI models so that predictions stay up-to-date.	High	5	8
US017	As an admin, I want to manage users so that system access is controlled.	Medium	5	3
US018	As a farmer, I want notifications for disease alerts so that I act on time.	Medium	5	3
US019	As a system, I want to support mobile devices so that farmers can use phones easily.	Medium	6	5
US020	As a system, I want offline support (PWA) so that farmers can use it in low network areas.	Medium	6	5
US021	As a farmer, I want crop calendar suggestions so that I plan farming activities.	Low	6	3
US022	As a farmer, I want to participate in a discussion forum so that I can share knowledge.	Low	7	3
US023	As an expert, I want to reply to farmer queries so that proper guidance is given.	Medium	7	3
US024	As a system, I want secure authentication so that user data is protected.	High	7	5
US025	As a system, I want cloud deployment so that the application is scalable and available.	High	8	8

8.4.2. Sprint Backlog

The Sprint Backlog is an extension of the Product Backlog that includes the chosen user stories, tasks and technical activities that the development team promises to accomplish within a particular sprint. It is the work to be implemented in a specific duration of time, normally one to four weeks, and developed at the sprint planning meeting.

Sprint Backlog is a short term implementation plan in Scrum methodology. It contains the high-priority user stories selected among the Product Backlog and their approximate effort (story points) and implementation information. Sprint backlog belongs to the development team and is regularly updated as the work is developed during the sprint.

In this project, Sprint Backlog is used to group user stories, including image upload, AI-based disease detection, treatment recommendation, analytics, and system deployment into several sprints. This method guarantees the gradual improvement, constant testing, and initial delivery of fundamental capabilities. With the help of a Sprint Backlog, the project is transparent and better time-managed, and it can constantly adjust itself on the feedback and performance assessment (Schwaber and Sutherland, 2020).

The Major Characteristics of a Sprint Backlog.

- Includes some user stories chosen in the Product Backlog.
- Time based on a definite sprint period.
- Effort estimation uses story points.
- Concentrates on performance and action.
- Change on a daily basis during the sprint.

Sprint 1 – User Access & Core Upload (Foundation Sprint)

Sprint Goal: Enable user registration, login, and image upload.

ID	User Story	Story Point
US001	Farmer registration	3
US002	Farmer login	2
US003	Upload crop leaf image	5
US004	Image preprocessing	5
US024	Secure authentication	5

In Sprint 1 we were working on laying down the main backbone of the system. The primary goal of this sprint was to allow access and basic system interaction without conflicts. In this sprint, we introduced user registration and login system and security authentication systems. We also designed crop image upload option where farmers can post the image of the crop leaves to be analyzed. Further, the process of image preprocessing was also introduced to achieve the desired

quality of uploaded images to guarantee the correct disease detection through AI. Every next system functionality was preconditioned by this sprint.

Sprint 2 – AI Disease Detection

Sprint Goal: Implement core AI prediction functionality.

ID	User Story	Story Point
US005	CNN-based disease classification.	8
US006	View prediction results	3
US007	View confidence score	2
US008	Treatment recommendations	5

Sprint 2 was supposed to bring in the essence of intelligence of the system. This sprint involved the incorporation of a Convolutional Neural Network (CNN) model to detect crop diseases using posted images. Other features we created are the ability of the farmers to see disease prediction outcomes and the confidence scores. Moreover, the treatment recommendations were worked out on the detected diseases. This sprint turned the system into a project that is an intelligent decision-support tool instead of being a simple application.

Sprint 3 – History & Recommendation Enhancement

Sprint Goal: Improve usability and monitoring.

ID	User Story	Story Point
US009	Organic treatment suggestions	3
US010	Pesticide & dosage guidance	5
US011	Store prediction history	5
US012	View prediction history	3

We improved the quality and utility of recommendations given to farmers in Sprint 3. The system was spread to provide organic and chemical treatment recommendations including the appropriate dosage. We also added an ability to save the history of predictions and provide the user with an opportunity to see their previous disease detection outcomes. This sprint enhanced the traceability and aided farmers in tracking crop health with time.

Sprint 4 – Analytics & Admin Control

Sprint Goal: Enable analytics and administrative control.

ID	User Story	Story Point
US013	Disease trend charts	3
US014	Analytics report generation	5
US015	Dataset management	8

Sprint 4 was concerned with data analysis and system management. In this sprint, we installed analytics dashboards, which show disease trends and visual reports. Functionalities like the administration of datasets were also built and enabled administrators to maintain and enhance quality training data. This sprint helped monitor, control, and make data-driven decisions in the system.

Sprint 5 – Model Optimization & Notifications

Sprint Goal: Improve AI accuracy and user engagement.

ID	User Story	Story Point
US016	AI model retraining	8
US017	Admin user management	3
US018	Disease alert notifications	3

Sprint 5 was devoted to enhancing the accuracy of the system and the interaction with the user. The retraining functionality of AI models was introduced to make sure that the disease detection model will be accurate with the arrival of new data. The system access controls were also added by administering the users of the administration. Also, notification of the farmers concerning the important cases of diseases was incorporated to ensure prompt intervention.

Sprint 6 – Mobile & Offline Support

Sprint Goal: Accessibility in rural environments.

ID	User Story	Story Point
US019	Mobile responsiveness	5
US020	PWA offline support	5
US021	Crop calendar suggestions	3

Sprint 6 was aimed at ensuring the system was available in rural and under-connectivity areas. In this sprint, we made the user interface friendly to the mobile device and added Progressive Web App (PWA) features that would allow the user to use it without an internet connection. Crop calendar recommendations were also brought to assist farmers take care of farming activities. This sprint this enhanced the ease of use and access.

Sprint 7 – Community & Expert Interaction

Sprint Goal: Knowledge sharing and expert advice.

ID	User Story	Story Point
US022	Farmer discussion forum	3
US023	Expert replies	3

In the 7th sprint, we were working on knowledge sharing and expert support. The discussion forum was introduced where farmers should post questions and experiences. Farmers were empowered to answer such questions and offer professional opinions. This sprint improved teamwork and created an encouraging agricultural community of the system.

Sprint 8 – Deployment

Sprint Goal: Make system production-ready.

ID	User Story	Story Point
US025	Cloud deployment	8

The last sprint which is Sprint 8 was devoted to the system deployment and readiness to the real-life application. This sprint saw the application moved to a cloud environment in order to guarantee scalability, availability and performance. The system was also tested, configured and optimized to finalize its functionality and non-functionality requirements. This sprint was the final stage of the development lifecycle.

8.5. Designs

8.5.1. Class Diagram

One of the most important objects in the design of objects is the class diagram. It gives a graphical depiction of the classes of the system, their attributes (data they store), methods (operations they perform) and their relationship with other classes. Basically it depicts the roadmap of the system, which is beneficial to the developers as it helps them comprehend how various parts of the system interact, the flow of data in the system, and the allocation of responsibilities. This way, it minimizes confusion in the development process and provides the system with a structure, maintenance and scalability (Shocodevin, 2025; Prepbytes, 2023).

- i. Basic Classes and their duties.

1. Crop Class:

Purpose: Represents individual crops submitted by farmers.

Key Attributes: cropName, growthStage, imageData, diseaseStatus.

Methods: uploadImage(), requestDiseaseDetection(), viewRecommendations().

Explanation: Centralizing all crop-related data and operations in this class ensures consistency. For example, when a farmer uploads a leaf image, the Crop class handles storing the image and initiating disease detection.

2. Farmer Class:

Purpose: Represents farmers who use the system.

Key Attributes: farmerID, name, location, contactInfo.

Methods: submitCrop(), viewPredictionHistory().

Explanation: This class links farmers with their crops and predictions. One farmer can submit multiple crops, creating a **one-to-many relationship**. This setup also allows tracking each farmer's interactions and history within the system.

3. AIService Class:

Purpose: Encapsulates the machine learning logic for disease detection.

Key Attributes: modelType, trainingData, accuracy.

Methods: analyzeCropImage(), generateRecommendation().

Explanation: By isolating AI logic in its own class, the system can update models or add new predictive features without affecting other parts of the system. When a Crop object requests disease detection, this class processes the data and returns a Prediction object.

4. Prediction Class:

Purpose: Stores AI-generated results.

Key Attributes: predictionID, diseaseDetected, confidenceLevel, timestamp.

Methods: storePrediction(), retrievePrediction().

Explanation: This class maintains a history of predictions, enabling farmers and experts to track trends over time. It also allows analytics to identify frequently occurring diseases or high-risk crops.

5. Admin Class:

Purpose: Manages the system and oversees operations.

Key Attributes: adminID, role, permissions.

Methods: manageUsers(), viewAnalytics(), updateSystemSettings().

Explanation: Admins interact with other classes indirectly by monitoring user activity, managing data integrity, and controlling system configurations.

The interaction among classes within the system captures the interactions among various components to bring out the intended functionality. A Farmer may post several Crops which forms a one to many relationship and each Crop passes through the AIService that creates a Prediction object, so that all the crop information is processed in an orderly manner and is stored as a reference point whenever required. Admins monitor all the communication and may request reporting or management of any of the classes. Class diagrams do not only depict relationship

diagrammatically, but also act as a guide to implementation in terms of how the code should be structured, how data travels between objects, and how the system can easily be extended, by adding new AI models or other types of users. Good relationships also enhance maintainability where modifications in a single class, including training of AI logic, does not affect other classes. As an example, when a farmer posts a new picture of the crop, the Crop object engages with the AIService to generate a Prediction object which is stored and connected to the farmer, and this illustrates how the object-oriented approach has grouped the operations together on logic, traceability and simplified the debugging process.

8.6. Screenshots

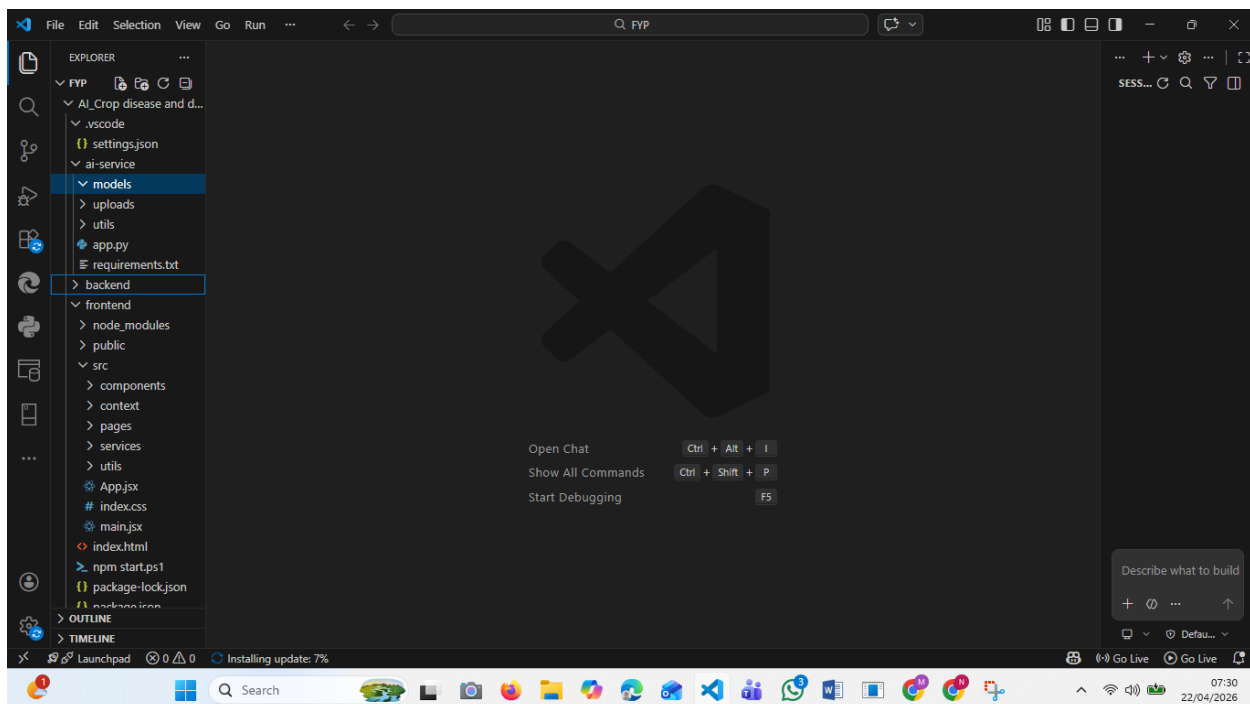


Figure 14: Proof 1

8.7. User Feedbacks

To effectively gauge the need of the system before the development of the system and to gauge the effectiveness of the system after its full implementation, user feedback was taken during two significant phases of the project, the pre-survey and the post-survey. This two-step feedback mechanism was essential in determining the user needs, justifying the system concept, and gauging the general success of the AI-Powered Crop Disease Detection and Recommendation System.

The pre-survey was carried out at the first phase of the project even before any development work was carried out. The primary aim of this survey was to establish the practical issues encountered by the users particularly farmers and those who are related to agriculture to detect and control the diseases of crops. It also sought to find out whether an AI-based solution was actually necessary. A total of 30 respondents were interviewed at this stage who included farmers and students, and other people who have an interest in the field of agriculture. The pre-survey findings clearly indicated that majority of the users were still relying on the conventional means to diagnose crop diseases. Such strategies involved physical inspection of crops, using the advice of agricultural specialists, and finding answers on the Internet using YouTube and Google.

But, users cited some drawbacks of these conventional methods. Many of the participants mentioned that visual identification is frequently not accurate since various diseases may present with the same symptoms. Some said that expert consultation is not readily available all the time and in rural settings where agricultural professionals might not be readily available. Also, searches over the internet were deemed to be time-consuming and occasionally confusing because the information out there on the internet was enormous and unverified. Many users also confessed that they are usually confused or uninformed when attempting to diagnose crop diseases themselves.

The next significant result of the pre-survey was the high level of interest towards technological solutions. Several users were willing to use a mobile or a web-based application that could identify crop diseases using pictures. They further highlighted that a system like that must be easy, quick and precise. This feedback was a solid reason why an AI-powered system needed to be developed and gave a clear guideline on what was required in the project. Thus, pre-survey served as a base to comprehend issues of the users and prove the relevance of the offered solution.

Once the system was fully developed and implemented a post-survey was carried out on the same group of users after interacting with the final application. During this step, users were able to test the system, uploading crop leaf images, being given a disease forecast, and accessing treatment and prevention advice. The key point of the post-survey was to assess whether the system effectively resolved the issues detected during the pre-survey and to quantify user satisfaction, usability, accuracy, and overall effectiveness.

The post-survey feedback was very positive. Users claimed that the system was user-friendly and no technical skills were needed. The easy-to-use interface, straightforward guidelines and rapid response time made the system easy to use even by people who are not very digitally literate. Majority of the users liked the image upload service and found it easy and effective.

Functionally, users were satisfied with the results of the disease detection. Numerous respondents mentioned that the system could identify diseases in crops and make useful predictions. The

confidence levels were included as it allowed the users to know the reliability of the results. The recommendation option was also well received because it gave really useful solutions like the treatment method, preventative measures, and overall advice when it comes to taking care of crops. This rendered the system more practical than the traditional ways of identification since it did not only identify the diseases but also recommended the next course of action to the user.

Users also indicated that using the system enhanced their knowledge about the diseases in crops. Quite a number of them said they were more confident about tackling crop related issues after using the application. One more valuable conclusion was that the system minimised the reliance on agricultural experts. This can be especially useful in isolated locations where professional assistance is scarce or absent.

Although there was a positive feedback, constructive suggestions on how it could be improved were also made by the users. There were suggestions to improve the user interface design to make it more modern and appealing. Some proposed to enhance the accuracy of detection of less common or rare diseases by increasing the size of the dataset to train the AI model. Some of the users also pointed out that the system needs an offline option so that it can be utilized in locations where there is low or no internet connection. Also, some minor technical problems like the occasional delays in loading were stated, which can be resolved in further versions.

In sum, a comparison of pre-survey and post-survey feedback shows clearly that the project was successful. The pre-survey proved that there were actual issues with the conventional ways of detecting crop diseases and that it was necessary to have an AI-based solution. The post-survey, however, demonstrated that the adopted system is a proper solution to these issues since it offers precise, quick and convenient detection of the disease and helpful advice. This evident user experience enhancement demonstrates that the AI-Powered Crop Disease Detection and Recommendation System is feasible, dependable, and of great value in real-life agricultural practices.

8.8. Future Work

8.8.1. Future Work Overview.

The AI-Powered Crop Disease Detection and Recommendation System is a system that has been developed to effectively detect the disease in crops with the help of deep learning and recommend users the appropriate treatment plan. The existing system is also able to detect disease based on images and provide valuable outputs to farmers. But agriculture is an ever-evolving discipline, and the needs of the users, climate, and disease dynamics evolve. Thus, a number of future betterment and additions can be implemented in order to make the system more developed, precise and universal.

The primary goal of future work is to enhance the performance of models, make them more accessible, and enhance capabilities of the system in order to be more useful in the real-life situations of farming.

8.8.2. Future Improvements.

Another of the most significant future enhancements is the increment of the AI model with the addition of the dataset. The system at present applies a small dataset to the Convolutional Neural Network (CNN) to train it. A bigger and more varied dataset with a variety of types of crops, stages of diseases, environment, and regional differences can be utilized in the future. This will assist the model to be more generalized and increase the accuracy of prediction particularly to rare or less common diseases. More sophisticated methods like transfer learning or hybrid deep learning models could also be used to enhance the performance and decrease training constraints.

The other significant enhancement is the creation of an offline version of the system. Currently, the system relies on internet connectivity to process and predict. But in most rural locations, there is little or slow internet. With the creation of a lightweight offline model or mobile-based AI integration, users can easily do basic disease-detecting tasks without having to be constantly connected to the internet. This will go a long way in enhancing accessibility and usability of the system by farmers in remote areas.

Real time environmental and weather information can also be incorporated into the system. The temperature, humidity, rainfall, and soil moisture in the agricultural conditions are significant in crop health. With the integration of these factors and the results of the disease detector, the system will be able to give more precise, contextual and preventive suggestions. This will bring the system closer to the real agricultural decision making and make it smarter.

The other significant enhancement to be made in future is the creation of a mobile version of the system. As the majority of farmers do not have access to computers, but use smartphones, a mobile-based application would make the system more feasible and accessible. Additional features like the ability to capture images on a real-time camera, use of voices to give instructions, and use of local languages can also be incorporated to enhance the usability by non-technical users.

Also, it is possible to expand the system by adding a farmer community and expert consultation module. This would enable users to provide information on their crop problems, talk over the problems, and get feedback on how the agricultural specialists or other seasoned farmers would resolve the issue. This system would enhance the knowledge exchange and establish an enabling agricultural system.

The other possible improvement is the establishment of a system of continuous learning. This would enable the AI model to be updated with new data and disease patterns on a regular basis. Consequently, the system will keep up-to-date and highly accurate overtime without the need to have full redevelopment.

Lastly, integration with IoT devices and drone-based monitoring can also be considered in the future. With the help of these technologies, farming is going to become more automated, efficient, and data-driven by enabling large-scale crop monitoring and early disease detection at the field level.

8.8.3. Conclusion

To summarize, the further evolution of this system has high chances to enhance its performance and use. Through these improvements, the system can be more precise, scalable, and accessible to more users. These enhancements will guarantee the further development of the AI-Powered Crop Disease Detection and Recommendation System as a stable and smart system to serve contemporary agriculture.